

FANUC Robot series
R-30iB MATE PLUS CONTROLLER
P-40iA Mechanical Unit
Operator's Manual

MAROP40IA07171E REV B

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FANUC America Corporation Patent List

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Conventions

 **WARNING**
Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

 **CAUTION**
Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

Note Information appearing next to NOTE concerns related information or useful hints

Safety

FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti–tie–down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.
- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end–use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.

- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro–magnetic interference (EMI), radio frequency interference (RFI), and electro–static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

Staying Safe While Teaching or Manually Operating the Robot

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.
- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 WARNING

Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.

- Using a low motion speed, single step the program for at least one full cycle.
- Using a low motion speed, test run the program continuously for at least one full cycle.

- Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

Staying Safe During Automatic Operation

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.
- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

Staying Safe During Inspection

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

Staying Safe During Maintenance

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 WARNING

Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 WARNING

Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

NOTE: Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today’s highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high–voltage electrostatic tools.

 **CAUTION**
Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.
- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.



WARNING

An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

Staying Safe While Operating the Paint Robot

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

Observe all safety rules and guidelines to avoid injury.

 WARNING

Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.

 WARNING

Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
 - A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
 2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
 3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
 4. When the dust has been removed, the covers must be replaced immediately.
 5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



WARNING

When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.
- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
 - Wear eye protection and protective gloves during application and removal.
 - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
 - If there is contact with the skin, wash with water.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
 - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
 - Adequate ventilation is required. Overexposure could cause drowsiness.
 - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
 - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.

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1 OVERVIEW

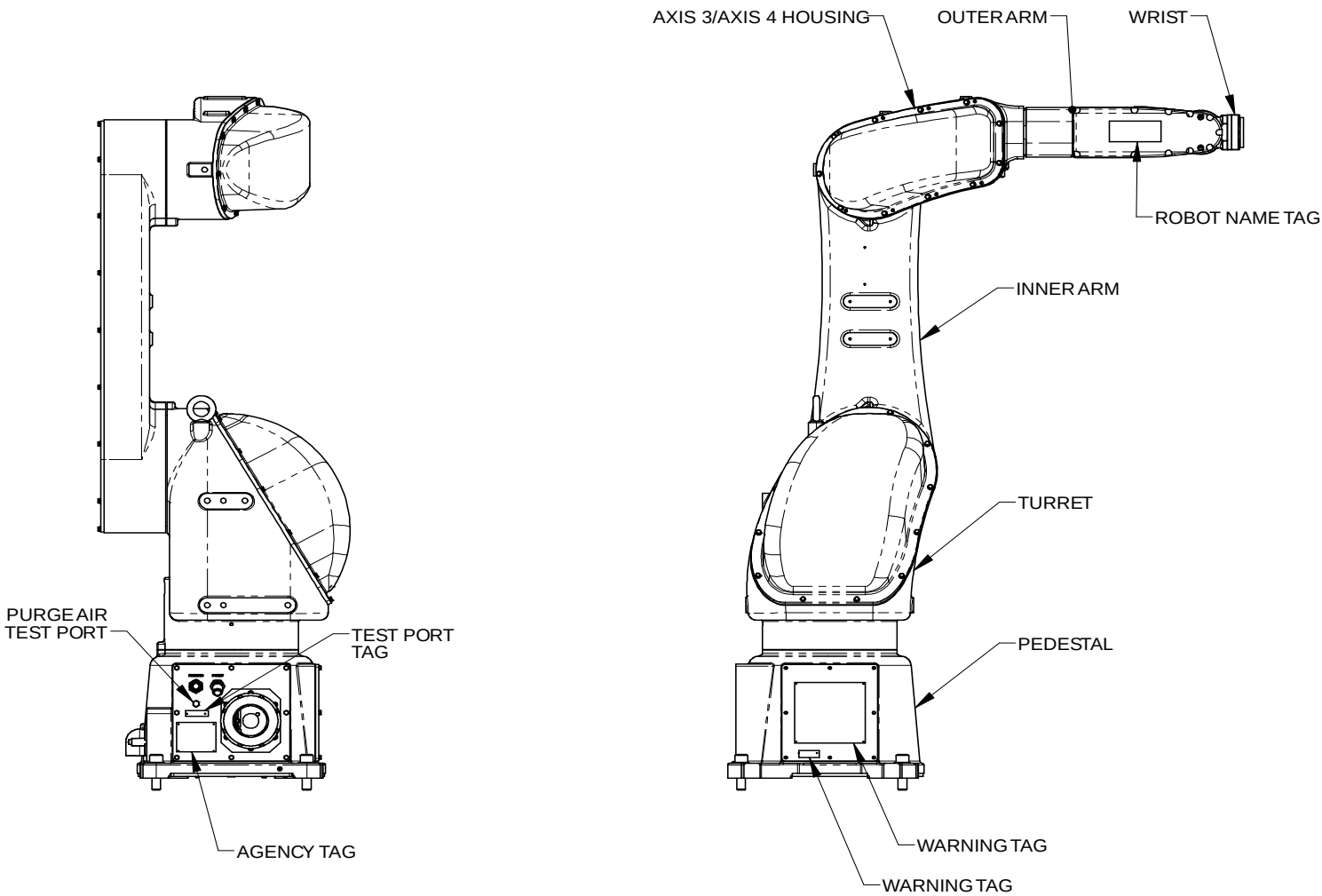
The P-40iA is a flexible mid-size painter robot:

- Compact arm design fits in small work areas
- Offers a 1300mm reach and 5kg payload
- Purged and pressurized arm is FM and ATEX Class Div. 1 approved
- Enclosed riser for cable connections and purge flow switch
- High-speed operation minimizes robot cycle times
- Upright, invert, angle or wall mounting accommodates a wide range of work areas
- 360 degrees rotation of J1 enables multi-directions operation

For detailed robot assembly information, see FANUC America Corporation *P-40iA Mechanical Unit Maintenance Manual*.

1.1 Major Components

Figure 1-1 DOC-7347-OVR-ASY, P-40iA MAJOR COMPONENTS





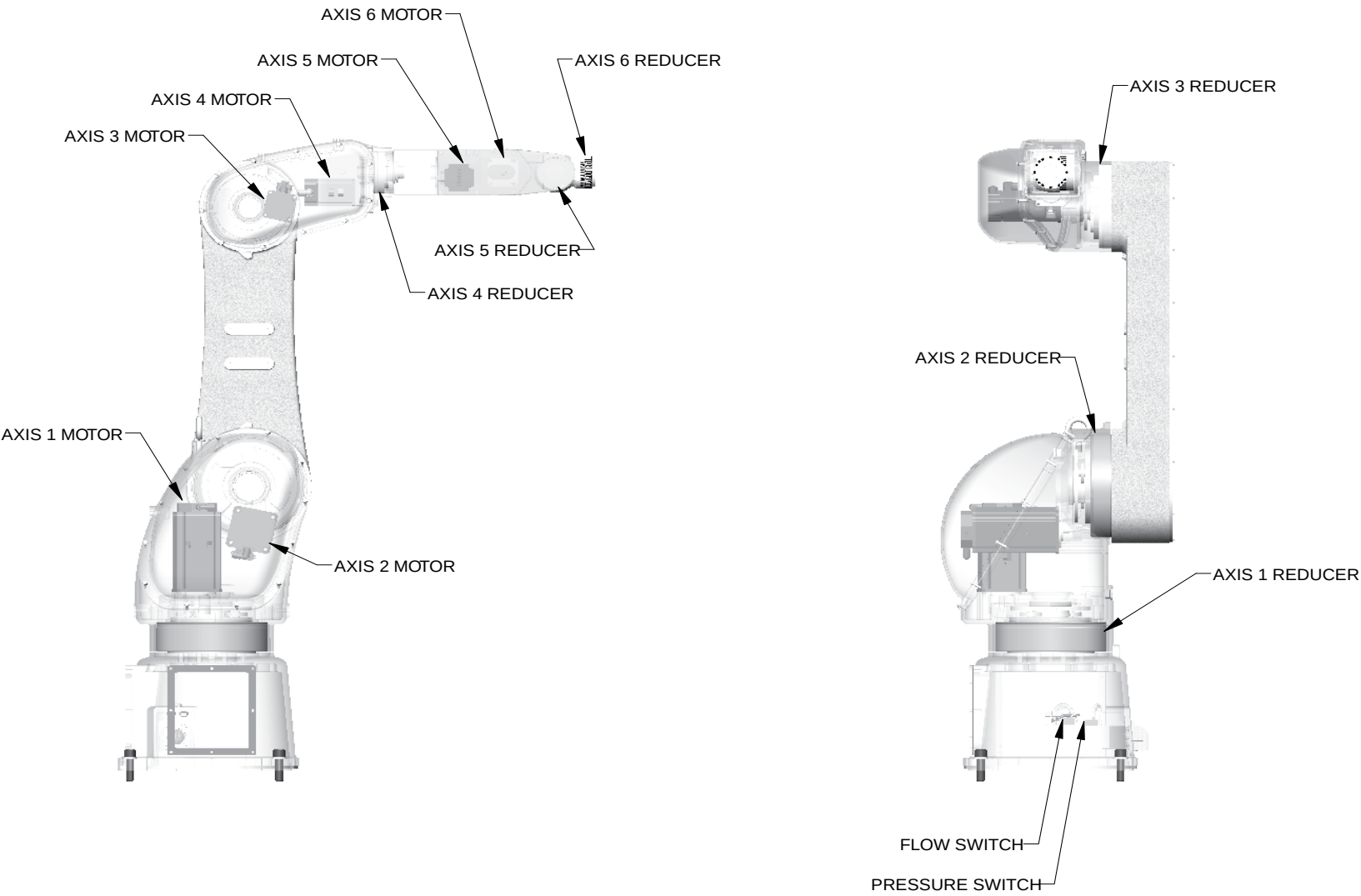
WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment



WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 1-2 DOC-7347-OVR-INT, P-40iA INTERNAL COMPONENTS



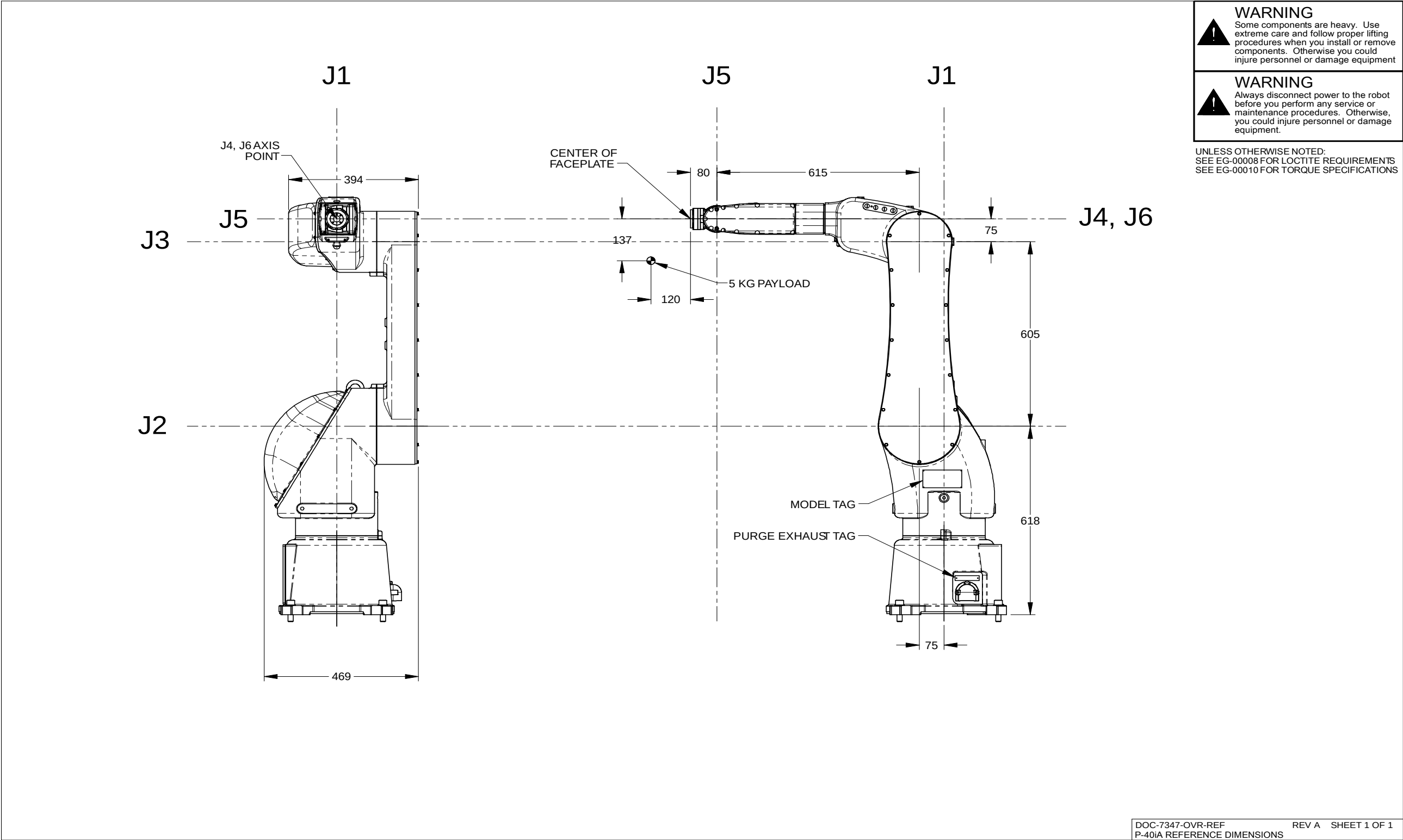
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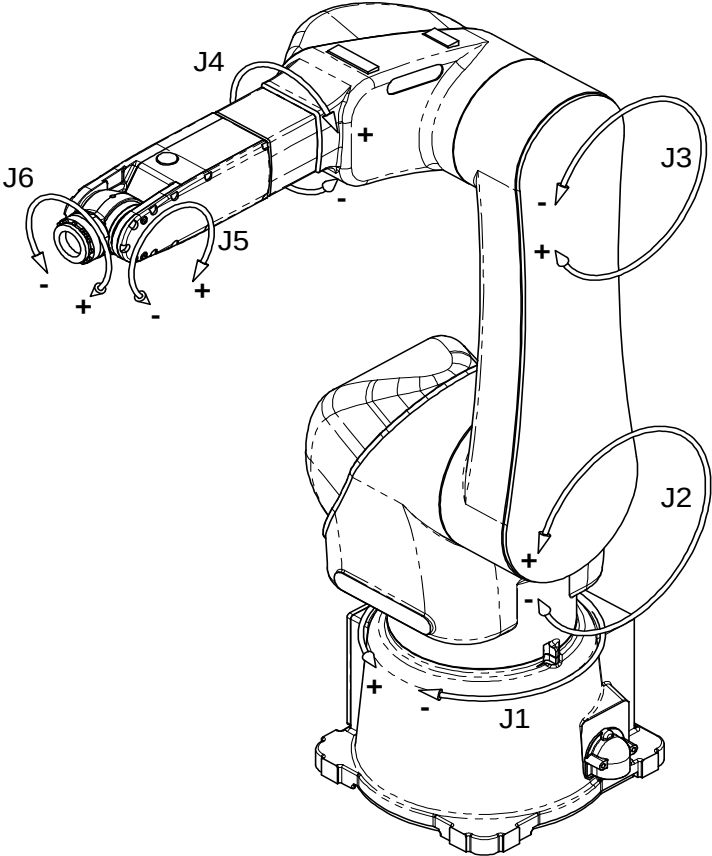
1.2 Reference Dimensions

Figure 1-3 DOC-7347-OVR-REF, P-40iA REFERENCE DIMENSIONS

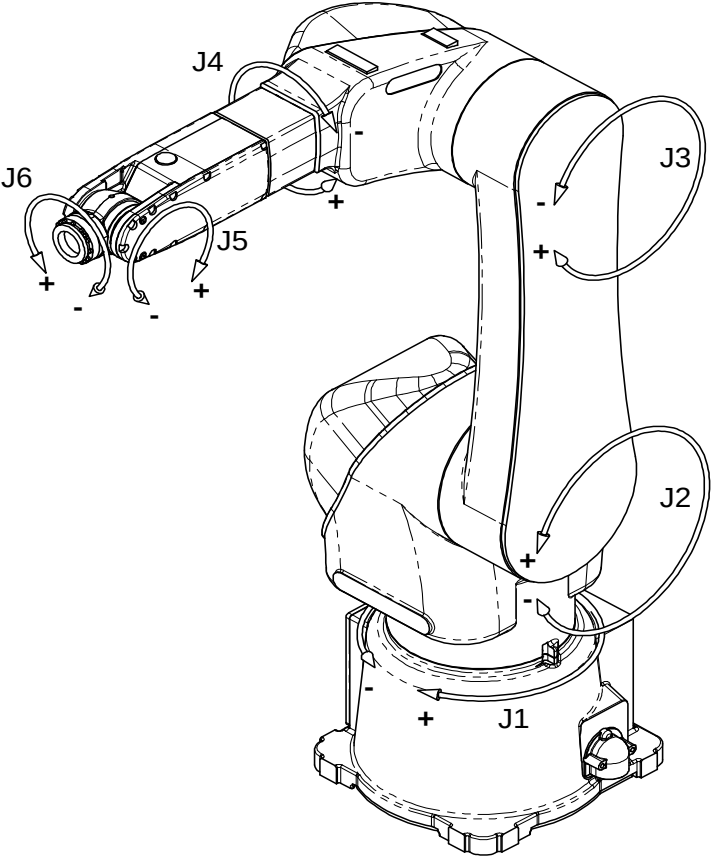


1.3 Configuration and Axis Direction

Figure 1-4 ET-7347-820-001, P-40iA AXIS DIRECTION



RIGHT HAND COORDIN ATE SYSTEM



LEFT HAND COORDINATE SYSTEM M

WARNING

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1.4 ET Drawing List

Figure 1-5 ET-7347-800-000, P-40iA ET DRAWING LIST



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SEE EG-00010 FOR TORQUE SPECIFICATIONS

P-40iA ET Drawing List	
Drawing	Description
ET-7347-800-000	ET Layout
ET-7347-800-001	Envelope
ET-7347-810-001	Mastering and Zero Position
ET-7347-820-001	Configuration and Axis Direction
ET-7347-830-001	Robot Mounting Interface and Loading
ET-7347-840-001	Robot Cable Penetration
ET-7347-850-001	Wrist Interface and Payload Information
ET-7347-850-002	Process Interface
ET-7347-860-001	Maintenance Details
ET-7347-860-002	Robot Maintenance
ET-7347-870-001	Lifting and Shipping
ET-7347-880-001	Robot Specifications and Requirements

2

SPECIFICATIONS

2.1 Robot Specifications and Requirements

Figure 2-1 ET-7347-880-001 Sheet 1 of 3, P-40iA - BASIC SPECIFICATIONS AND REQUIREMENTS



WARNING
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WARNING
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UNLESS OTHERWISE NOTED:
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SEE EG-00010 FOR TORQUE SPECIFICATIONS

BASIC SPECIFICATIONS AND REQUIREMENTS										
		GENERAL ENVIRONMENTAL REQUIREMENTS							GENERAL POWER SUPPLY RE Q	GENERAL AIR SUPPLY REQ .
		OPERATING TEMP.		STORAGE & SHIPPING TEMP.		HUMIDITY	ATMOSPHERE	VIBRATION		
		°C	°F	°C	°F	%				
ROBOT	MAX VALUE	45	113	60	140	95 (NOT CONDENSING)	FREE OF CORROSIVE GASES	LESS THAN 0.5 G	APPROPRIATE CONSTANT POWER SOURCE (UNIVERSAL VOLTAGES) 200, +10/-15%, 50/60HZ, 3 PHASE. SOURCE TO BE WITHIN 30.5M (100 FEET) OF REQ LOCATION	CLEAN , DRY AIR SUPPLY: 100 PSI GAGE, FOR FLOW RATE - SEE ET-7347-880-001; FILTRATION -5 MICRONS ABSOLUTE. SOURCE TO BE WITHIN 15.2M (50 FEET) OF REQ LOCATION
		MIN VALUE	10	50	-10	14				
CONTROLLER	MAX VALUE	45	113	60	140	95 (NOT CONDENSING)				
	MIN VALUE	0	32	-20	-4					

Figure 2-2 ET-7347-880-001 Sheet 2 of 3, P-40iA - BASIC SPECIFICATIONS AND REQUIREMENTS

ROBOT PURGE AIR SUPPLY

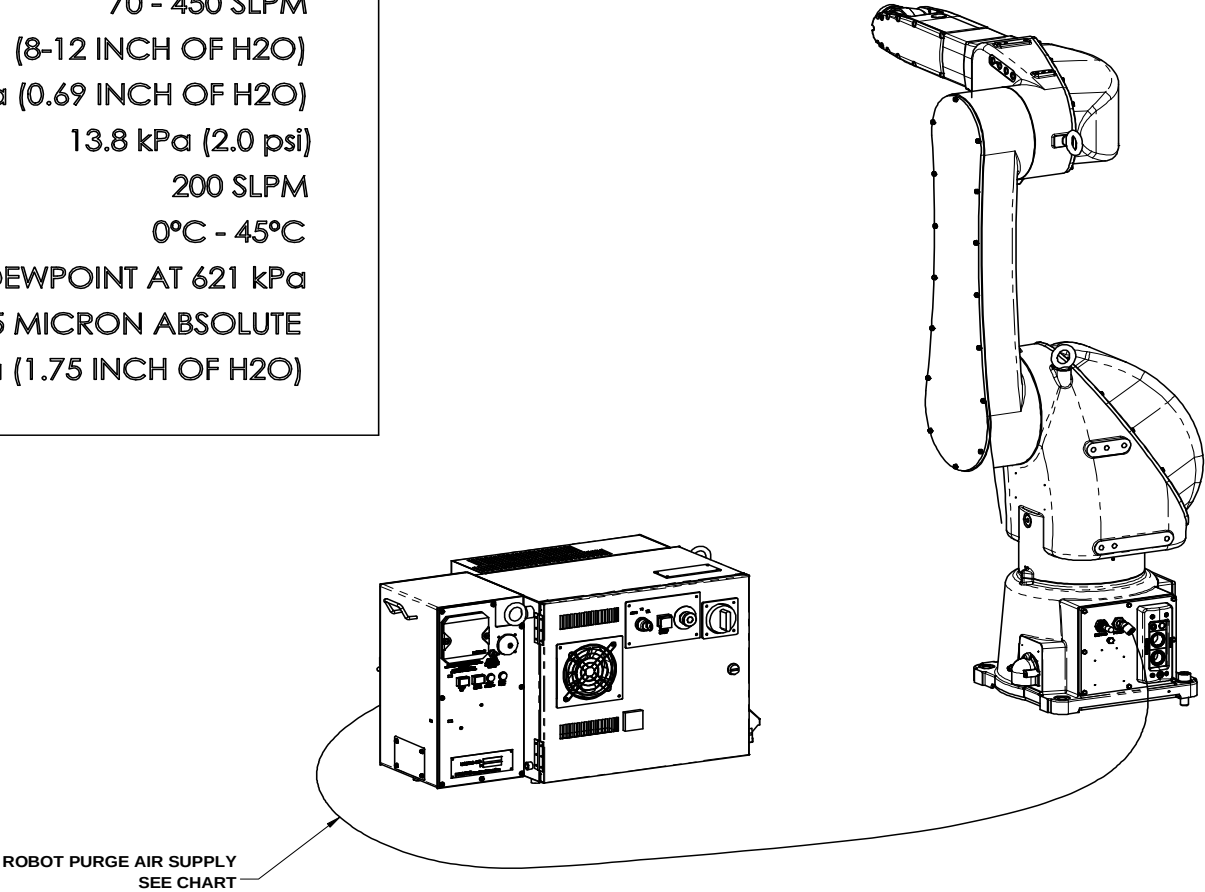
MINIMUM FLOWSWITCH ACTIVATION	566 SLPM
MAXIMUM FLOWSWITCH ACTIVATION	795 SLPM
AIR SUPPLY PRESSURE	448 - 517 kPa (65 - 75 psi) DYNAMIC
MAXIMUM AIR SUPPLY TEMPERATURE	45°C
MAXIMUM MAINTENANCE FLOW	70 - 450 SLPM
MAINTENANCE PRESSURE	1.993 - 2.986 kPa (8-12 INCH OF H2O)
MINIMUM OVERPRESSURE	0.172 kPa (0.69 INCH OF H2O)
MAXIMUM OVERPRESSURE	13.8 kPa (2.0 psi)
MAXIMUM LEAKAGE FLOW	200 SLPM
OPERATING AMBIENT TEMPERATURE	0°C - 45°C
HUMIDITY	-40°C PRESSURE DEWPOINT AT 621 kPa
FILTRATION	5 MICRON ABSOLUTE
EXHAUST PRESSURE	0.44 kPa (1.75 INCH OF H2O)

EO-7347-150-004

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UNLESS OTHERWISE NOTED:
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SEE EG-00010 FOR TORQUE SPECIFICATIONS



ROBOT PURGE SUPPLY LENGTH	# of 12mm SUPPLY LINES
UP TO 15m	1
OVER 15m	2

2.2 Envelope

Figure 2-4 ET-7347-800-001 Sheet 1 of 2, P-40iA ENVELOPE

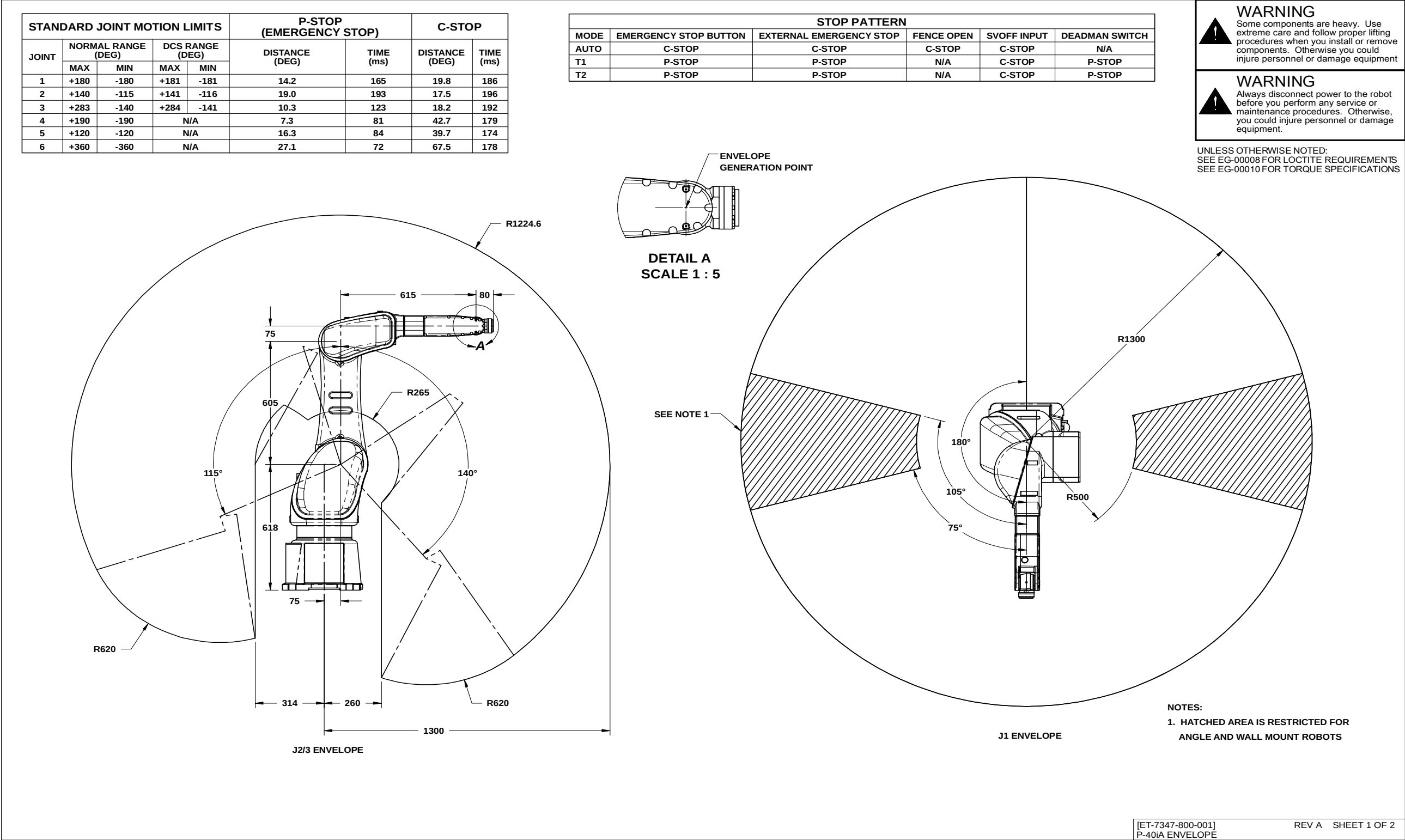
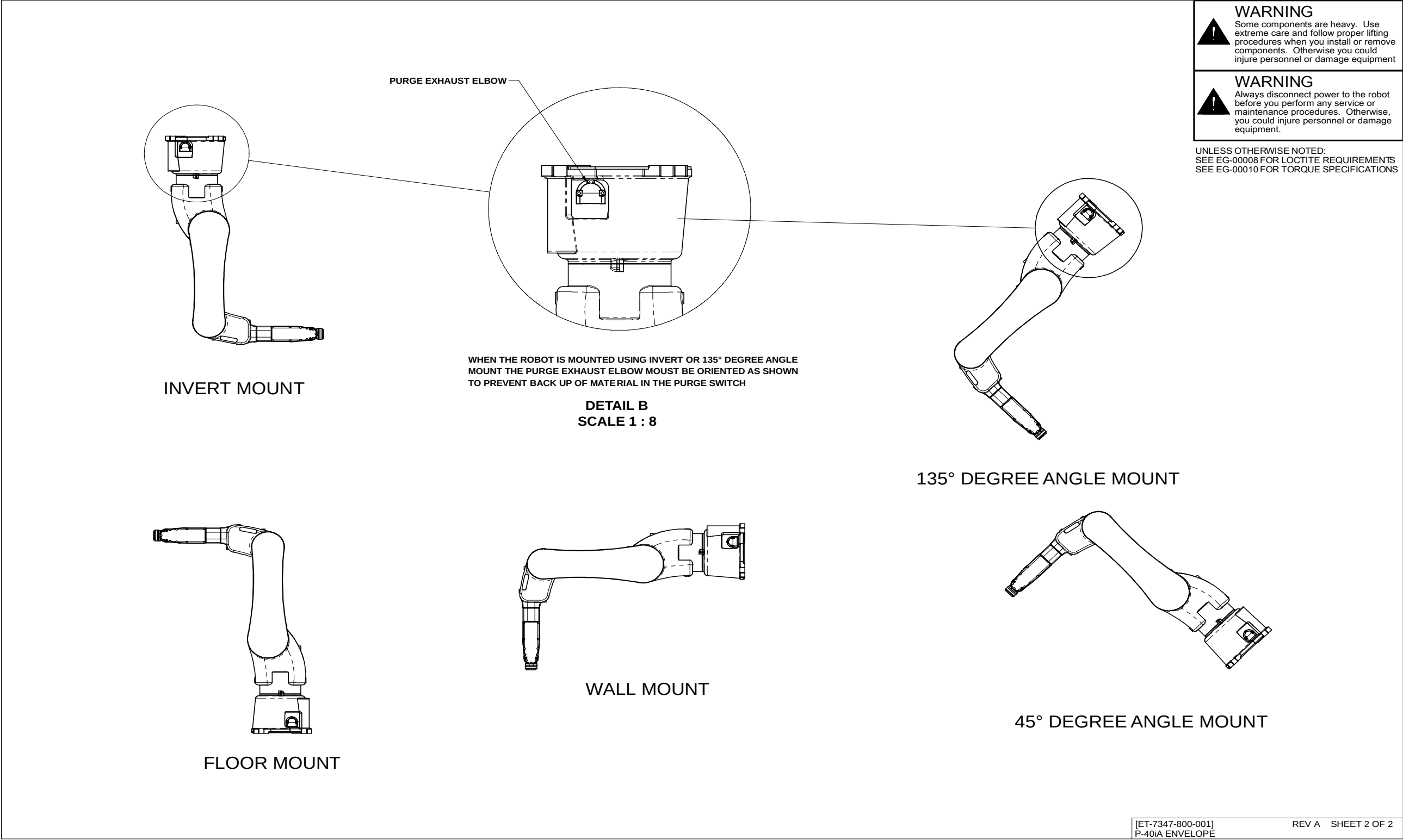


Figure 2-5 ET-7347-800-001 Sheet 2 of 2, P-40iA ENVELOPE



3

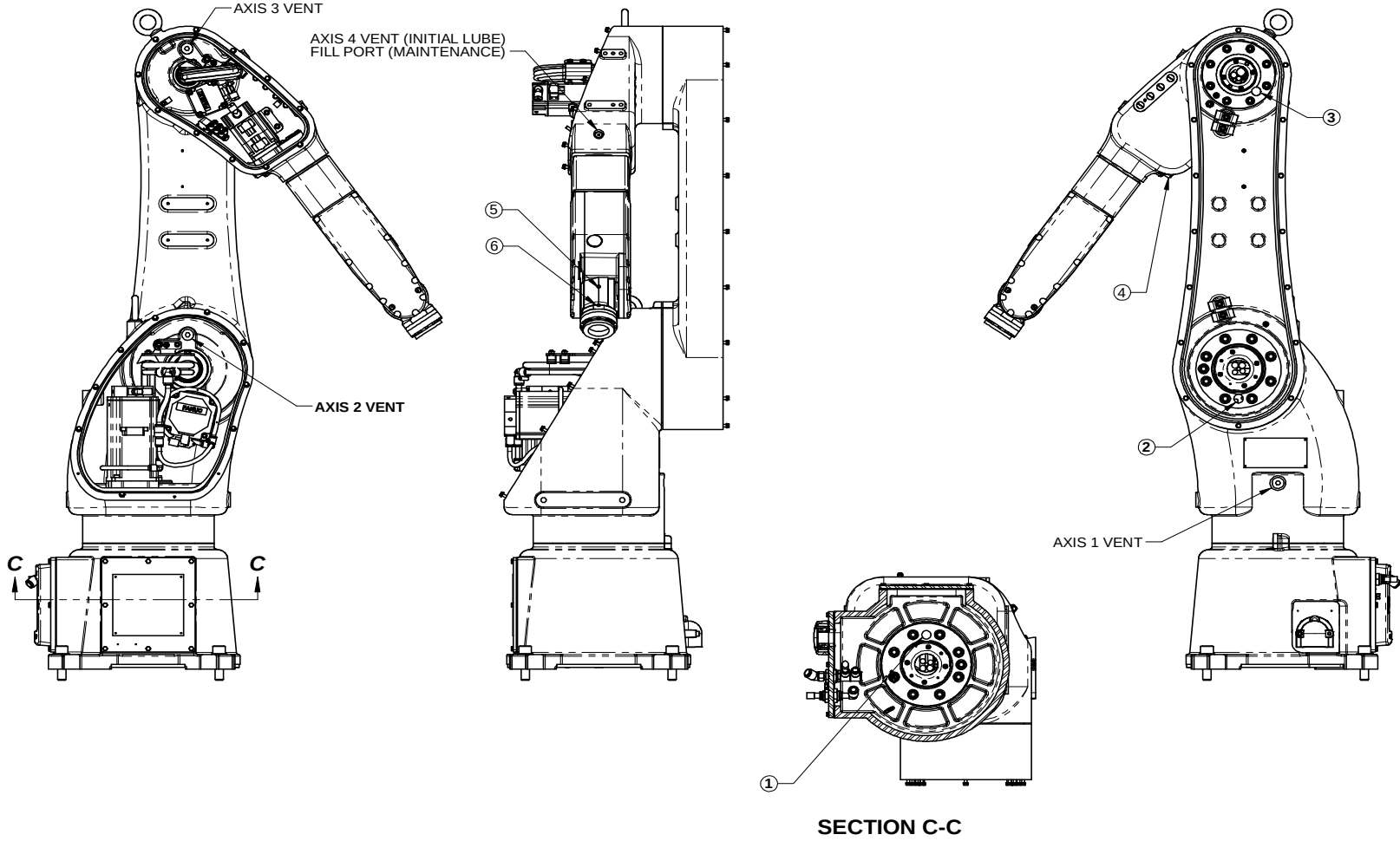
PREVENTATIVE MAINTENANCE

3.1 Maintenance Details

Figure 3-1 ET-7347-860-001 Sheet 1 of 2, LUBE PORTS AND MAINTENANCE

LUBE REQUIREMENT					
LUBE POINT	ASSEMBLY LOCATION	GREASE TYPE	INITIAL AMOUNT	RE-LUBE AMOUNT	FREQUENCY (HRS)
1	AXIS 1 RV	VIGO GREASE RE NO. 0	940ml (846g)	940ml (846g)	11,520
2	AXIS 2 RV	VIGO GREASE RE NO. 0	930ml (837g)	930ml (837g)	11,520
3	AXIS 3 RV	VIGO GREASE RE NO. 0	530ml (477g)	530ml (477g)	11,520
4	AXIS 4 HARMONIC REDUCER	GREASE 4B NO. 2	15ml (13.5g)	2ml (1.8g)	15,360
5	AXIS 5 HARMONIC REDUCER	GREASE 4B NO. 2	N/A	2ml (1.8g)	15,360
6	AXIS 6 HARMONIC REDUCER	GREASE 4B NO. 2	N/A	2ml (1.8g)	15,360

STD. FANUC PART NO.	ZINC FREE FANUC PART NO.	GREASE TYPE - SIZE
XGMF-16799	N/A	VIGO GREASE RE NO. 0 - 2 KG
XGMF-16156	N/A	VIGO GREASE RE NO. 0 - 16 KG
XGMF-20526	N/A	GREASE 4B NO. 2 - 0.5 KG
A05B-1139-K021	N/A	GREASE 4B NO. 2 GREASING KIT (TUBE OF GREASE PLUS INJECTOR)



- 1. INITIAL & REBUILD LUBE DONE FROM POSITION SHOWN BELOW (COVERS REMOVED FOR CLARITY).
- 2. LUBE WITH CALIBRATED GREASE PUMP PER SPECIFICATION. AXIS 5 AND 6 HAVE BEEN GREASED BEFORE SHIPPING AND DO NOT REQUIRE INITIAL LUBRICATION.
- 3. REMOVE PLUGS (LUBE & VENT) BEFORE LUBE. REINSTALL PLUGS WHEN LUBE IS COMPLETE. USE LOCTITE 567 THREAD SEALANT (HDWMO0000031850) ON AXIS 1, 2, AND 3 FILL PORTS. APPLY A LIGHT COATING OF GREASE TO ORING OF AXIS 4 FILL PORT AND AXIS 1, 2, 3, AND 4 VENT PORTS.
- 4. TIP PRESSURE OF GREASE GUN MUST NOT EXCEED 0.1 MPa WHEN GREASING AXIS 1, 2, 3, and 4.
- 5. USE EO-6414-703-000 TO FILL AXIS 1, 2, AND 3.

WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment

WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

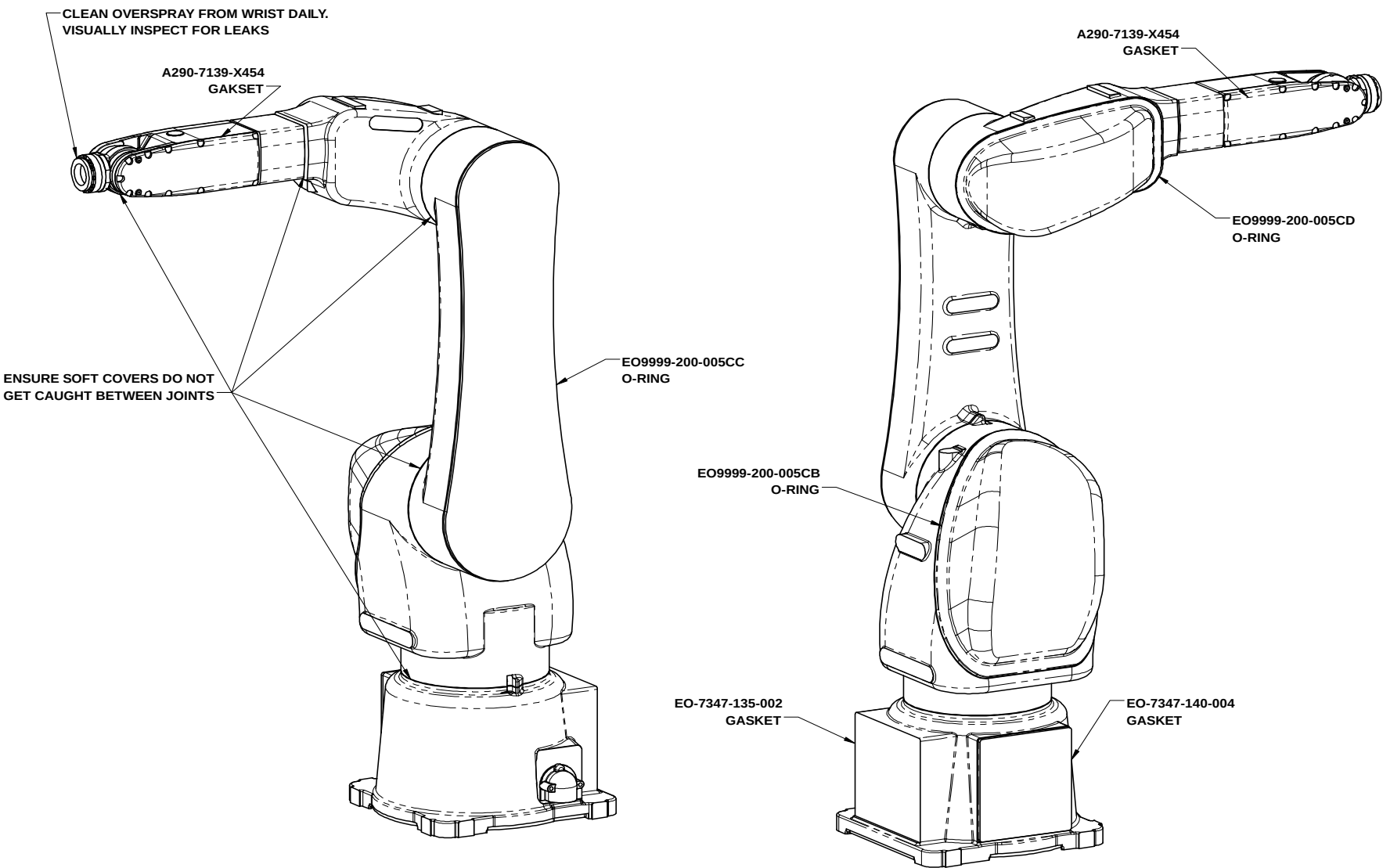
AXIS 1, 2, AND 3 RE-LUBE PROCEDURE

- 1. POSITION THE ROBOT IN THE GREASING POSITION SHOWN (COVERS REMOVED FOR CLARITY).
- 2. REMOVE THE VENT PORT AND SUPPLY NEW GREASE THRU THE FILL PORT PER THE CHART. OLD GREASE WILL BE PUMPED OUT OF THE VENT PORT. USE TOOL EO-7347-725-000 TO CONTAIN OLD GREASE.
- 3. REINSTALL PLUGS WHEN LUBE IS COMPLETE. USE LOCTITE 567 THREAD SEALANT (HDWMO0000031850) ON AXIS 1, 2, AND 3 FILL PORTS. APPLY A LIGHT COATING OF GREASE TO ORING OF AXIS 4 FILL PORT.
- 4. AFTER GREASING, OPERATE THE ROBOT THRU 60 DEGREES RANGE OF MOTION OR MORE FOR A MINIMUM OF 20 MINUTES.
- 5. REMOVE THE VENT PLUG TO RELEASE ANY EXCESS PRESSURE. SOME GREASE MAY LEAK DURING THIS PROCES S. BE SURE TO POSITION THE ROBOT TO MINIMIZE GREASE LEAKAGE. WIPE AWAY EXCESS GREASE.
- 6. REATTACH THE VENT PORTS AND LU BRICATE THE O-RING WITH GREASE.

AXIS 4, 5, AND 6 RE-LUBE PROCEDURE

- 1. REMOVE AXIS 4 MAINTENANCE PLUG, AXIS 5 SEAL BOLT, A ND AXIS 6 SEAL BOLT.
- 2. SUPPLY REGULATED AMOUNT OF GREASE USING THE INJECTION SYRINGE. **NOTE:** SOME GREASE MAY LEAK DURING GREASING OR IMMEDIATELY AFTER GREASE HAS BEEN SUPPLIED. EVEN IN THIS CASE, DO NOT SUPPLY GREASE BEYOND THE SPECIFIED AMOUNT.
- 3. REPLACE AXIS 4 MAINTENANCE PLUG, AXIS 5 SEAL BOLT, A ND AXIS 6 SEAL BOLT. **NOTE:** WHEN REUSING A SEAL BOLT, SEAL TAPE MUST BE APPLIED.

Figure 3-2 ET-7347-860-001 Sheet 2 of 2, LUBE PORTS AND MAINTENANCE





WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment



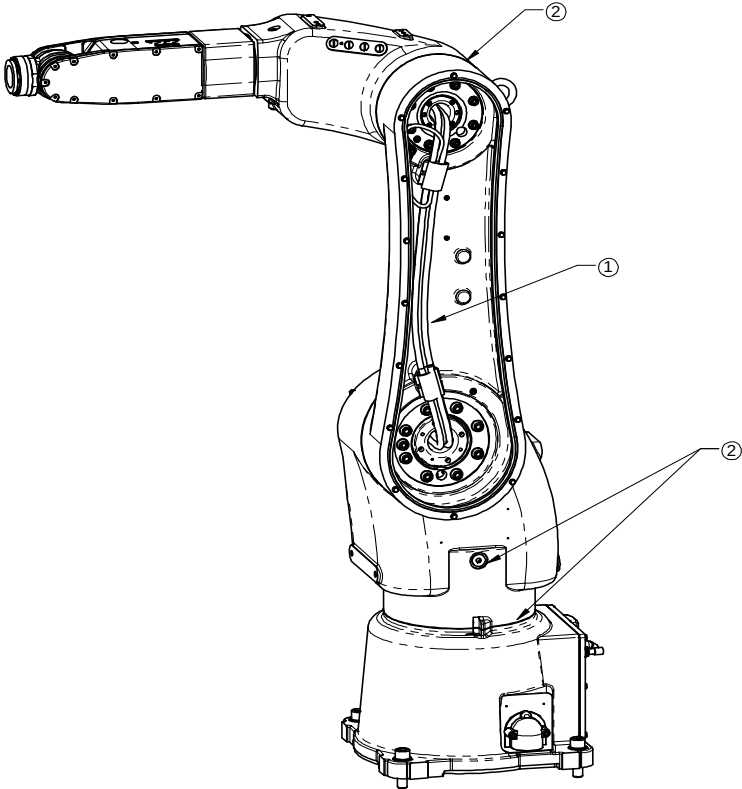
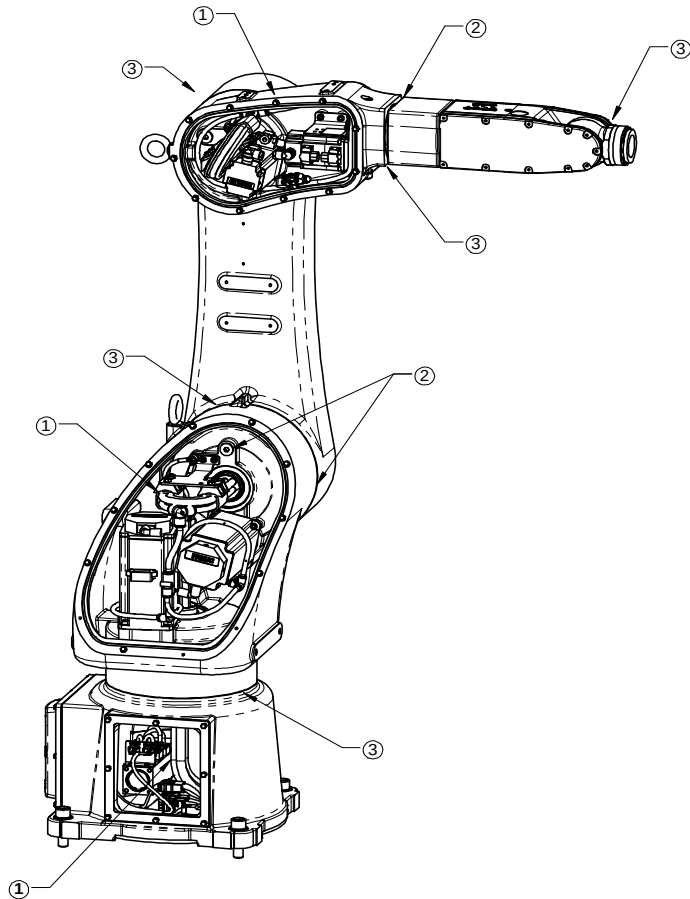
WARNING
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UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

CONSUMABLE ITEMS	
COVER SCREWS	SHHMM005080020S & SHHMM005080016S
LOCK WASHERS	WSHLM050092012S
FLAT WAHSE R S	WSHFM050100010O
HD CABLE TIE	TYRPO000000002O
LUBE PLUGS	PLGME000000139S & PLGME075016000S

3.2 Robot Maintenance

Figure 3-3 ET-7347-860-002, ROBOT MAINTENANCE





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WARNING
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UNLESS OTHERWISE NOTED:
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SEE EG-00010 FOR TORQUE SPECIFICATIONS

No.	DESCRIPTION	MAINTENANCEINTERVAL
1	CHECK INTERNAL/EXTERNAL CABLES	8000 HOURS
2	VISUAL CHECK FOR INTERNAL/EXTERNAL GREASE LEAKS	8000 HOURS
3	INSPECT ROBOT SOFT COVERS ARE NOT IN DYNAMIC JOINT	DAILY

3.3 Robot Cleaning and Soft Covers

Paint Robot Cleaning:

- A soft paint brush should be used on the seals to clean the paint from them without damaging the seals.
- The wrist/applicator may be cleaned using a lint-free rag soaked with solvent.
- After cleaning, seals of the wrist can be coated with petroleum jelly. This keeps paint from sticking to the seal and causing excessive friction in the wrist joint, which can cause collision guard or overcurrent errors.
- Cables in cable carrier can be cleaned with a lint-free rag soaked with solvent. After cleaning, the cables can be coated with a layer of petroleum jelly. This keeps paint from sticking and will make it easier to clean the cables in the future.
- The front of a rail can be coated with petroleum jelly. This will make it easier to remove paint.

Robot Cleaning for Dusty Applications:

- Dust may be cleaned from the joints of the robot using a lint-free rag or soft paint brush.
- Where allowed, a vacuum can be used to clean around robot joints.
- For dusty applications, the use of petroleum jelly on seals is not recommended because it will attract and retain the dust.

Purge Exhaust

If paint is observed accumulating inside the purge exhaust, the following should be considered:

- Modify the paint path so the robot doesn’t paint itself.
- Correct the balance of the booth, so paint isn’t sucked past the exhaust.

Paint Robot Soft Covers:

- FANUC America Corporation recommends the use of fabric covers with painting robots that are used in painting or dusty applications.
- Most robot models have custom fabric covers that have been validated by FANUC America Corporation.
- Failure to use robot covers may allow excessive paint build-up or dust accumulation on the seals. Over time, this may cause the robot to become inoperable.
- Both re-usable and disposable covers are available.
- Robot covers should never pull the process routing toward the robot arms. Tight covers can cause premature wear and failure of the process lines due to abrasion between the robot arm and process lines. Tight covers can also cause process line kinking and failure.
- FANUC America Corporation suggests the following typical maintenance schedule, but it is only a guideline:

Cover	Maintenance Frequency
Robot Covers	Monthly
Arm Covers	Weekly
Wrist Covers	Daily

4

INSTALLATION

4.1 Robot Cable Penetration

Figure 4-1 ET-7347-840-001 Sheet 1 of 4, P-40iA CABLE PENETRATION

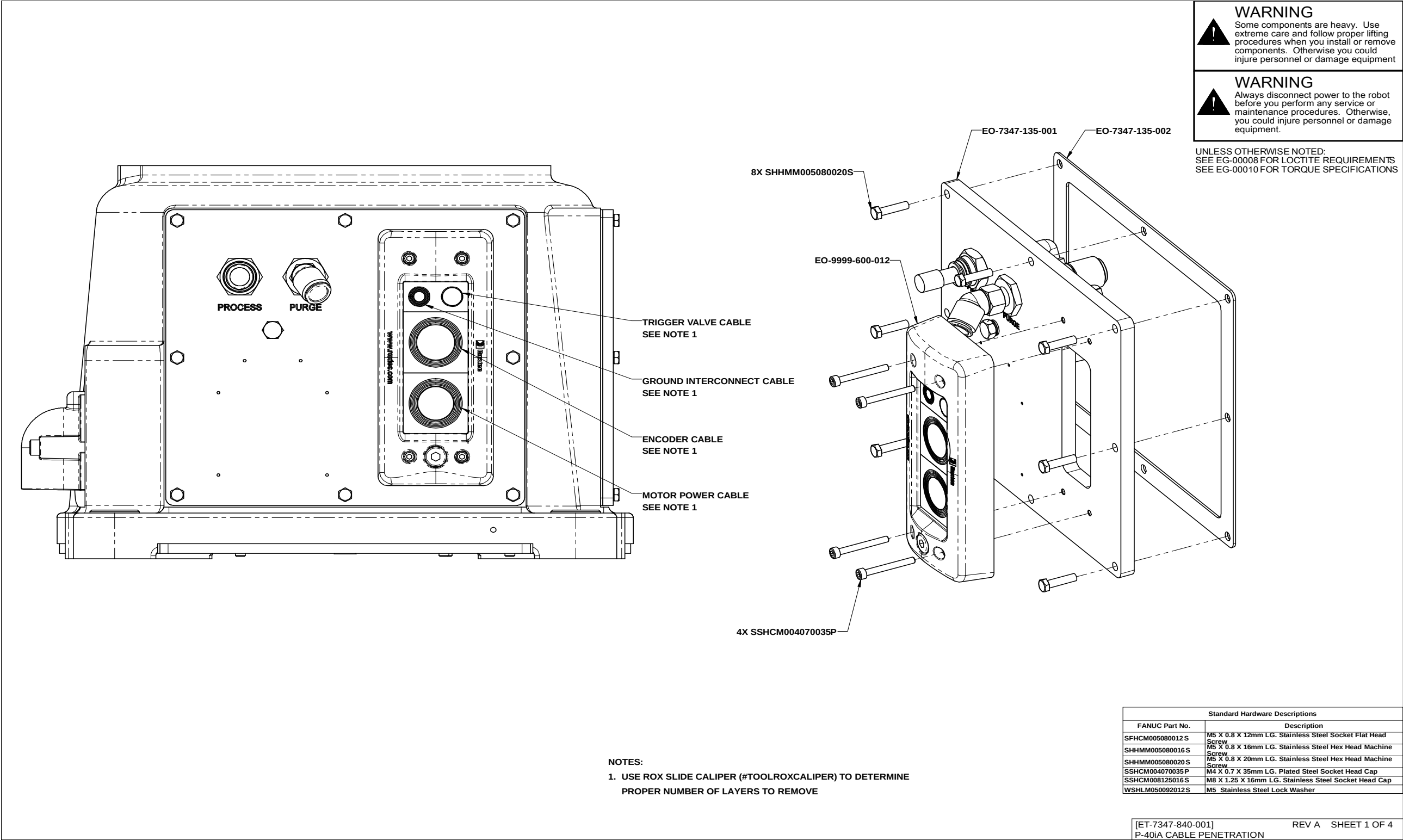


Figure 4-2 ET-7347-840-001 Sheet 2 of 4, P-40iA CABLE PENETRATION

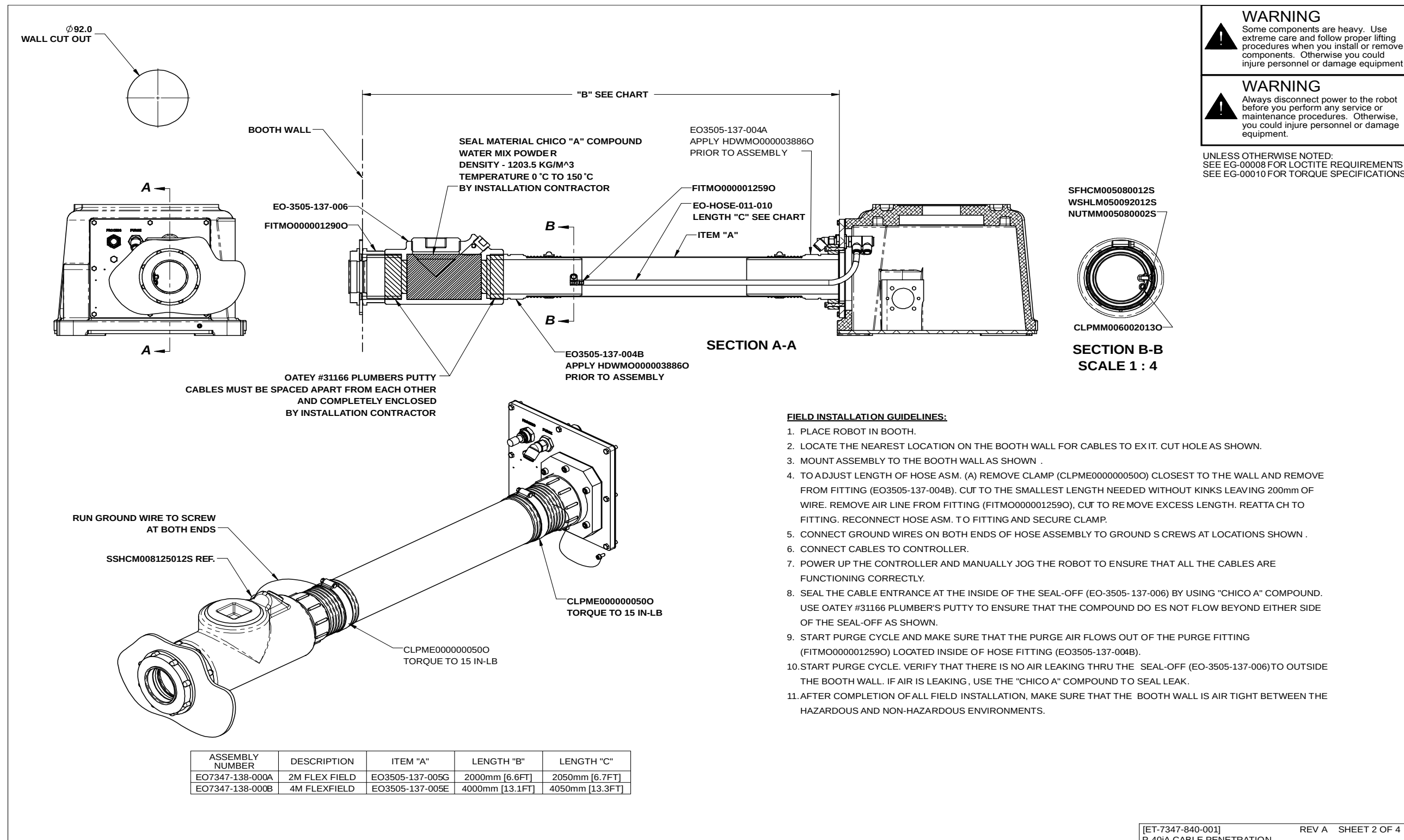


Figure 4-3 ET-7347-840-001 Sheet 3 of 4, P-40iA CABLE PENETRATION

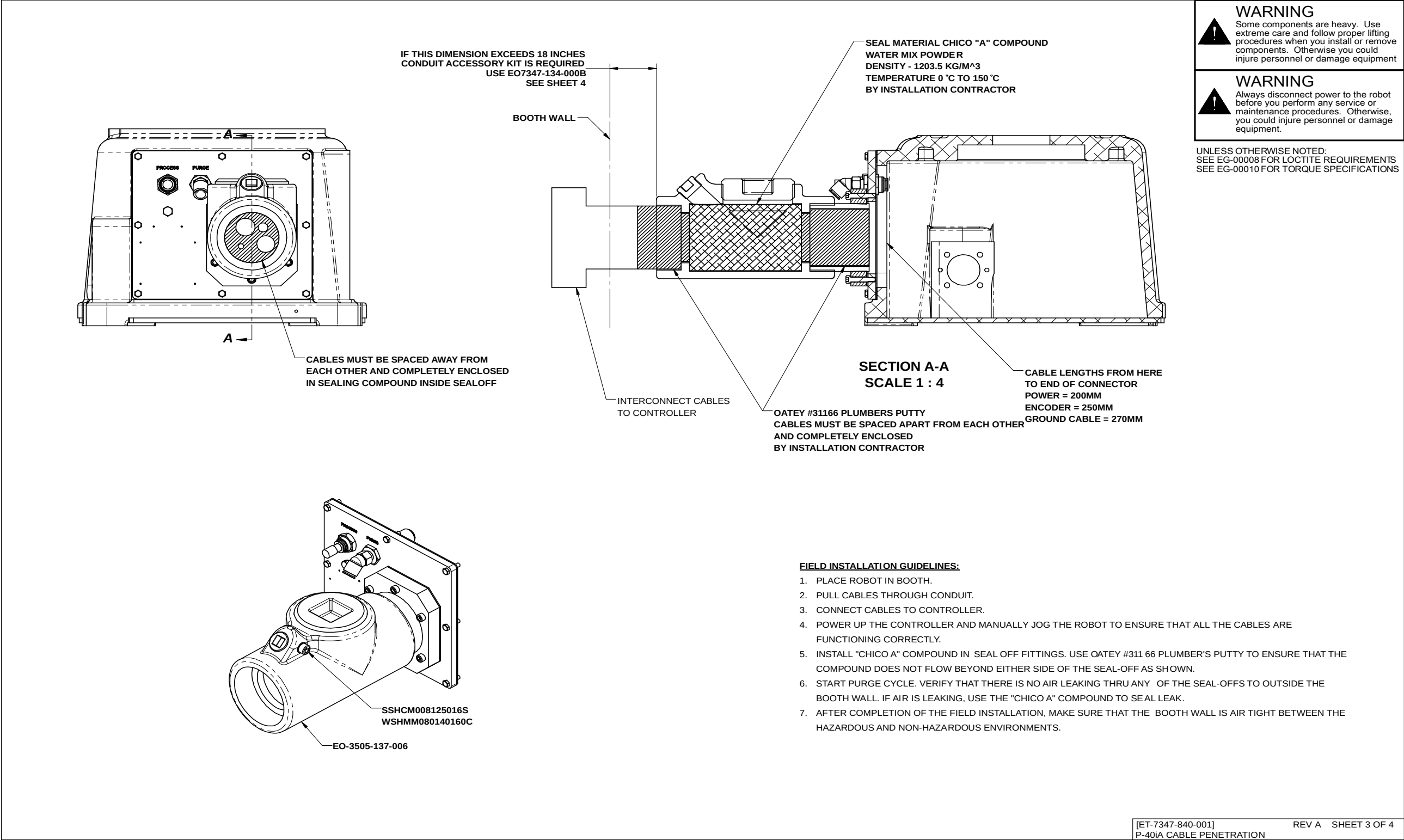
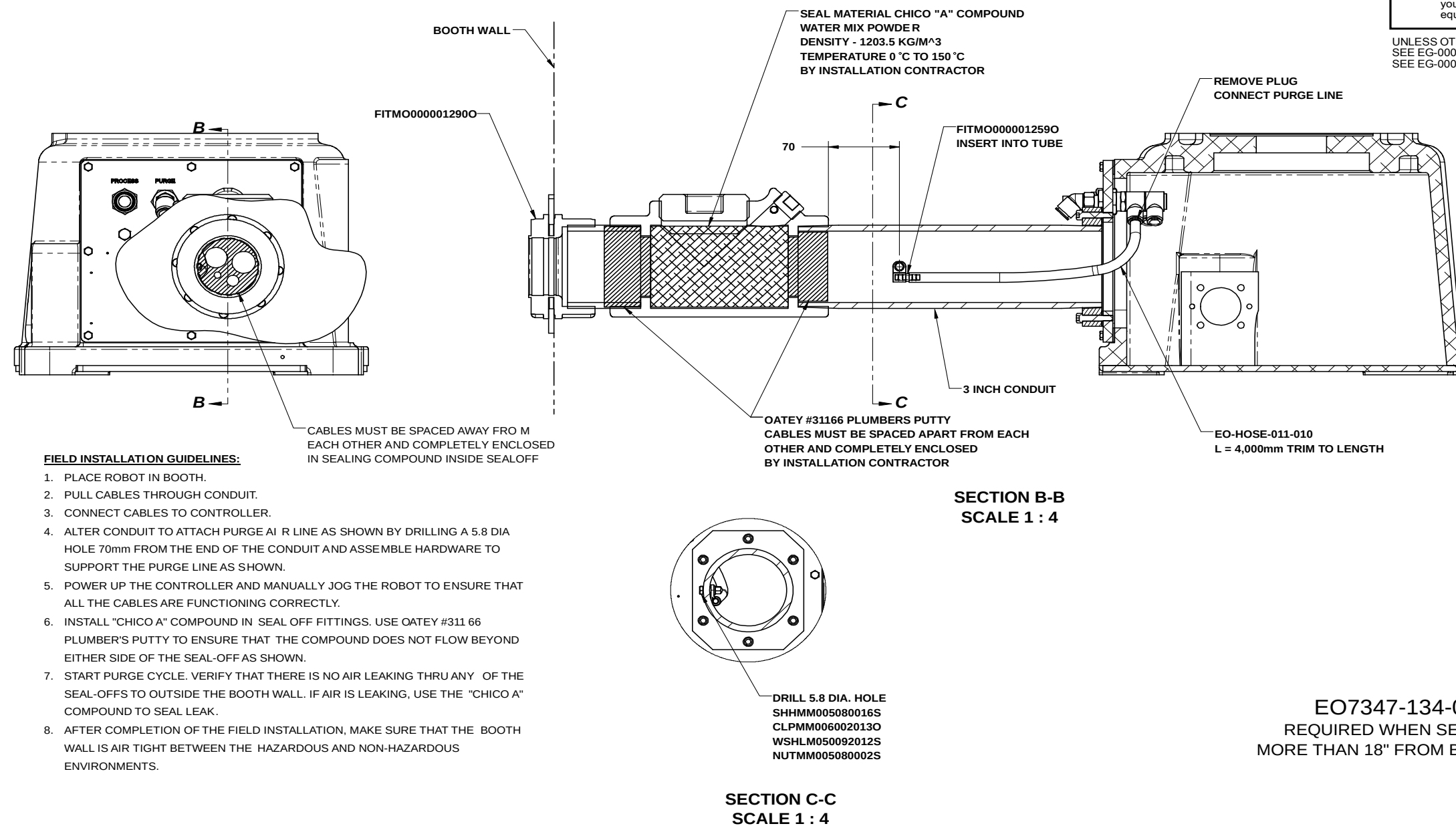


Figure 4-4 ET-7347-840-001 Sheet 4 of 4, P-40iA CABLE PENETRATION



WARNING

Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment

WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

FIELD INSTALLATION GUIDELINES:

1. PLACE ROBOT IN BOOTH.
2. PULL CABLES THROUGH CONDUIT.
3. CONNECT CABLES TO CONTROLLER.
4. ALTER CONDUIT TO ATTACH PURGE AIR LINE AS SHOWN BY DRILLING A 5/8 DIA HOLE 70mm FROM THE END OF THE CONDUIT AND ASSEMBLE HARDWARE TO SUPPORT THE PURGE LINE AS SHOWN.
5. POWER UP THE CONTROLLER AND MANUALLY JOG THE ROBOT TO ENSURE THAT ALL THE CABLES ARE FUNCTIONING CORRECTLY.
6. INSTALL "CHICO A" COMPOUND IN SEAL OFF FITTINGS. USE OATEY #311 66 PLUMBER'S PUTTY TO ENSURE THAT THE COMPOUND DOES NOT FLOW BEYOND EITHER SIDE OF THE SEAL-OFF AS SHOWN.
7. START PURGE CYCLE. VERIFY THAT THERE IS NO AIR LEAKING THRU ANY OF THE SEAL-OFFS TO OUTSIDE THE BOOTH WALL. IF AIR IS LEAKING, USE THE "CHICO A" COMPOUND TO SEAL LEAK.
8. AFTER COMPLETION OF THE FIELD INSTALLATION, MAKE SURE THAT THE BOOTH WALL IS AIR TIGHT BETWEEN THE HAZARDOUS AND NON-HAZARDOUS ENVIRONMENTS.

SEAL MATERIAL CHICO "A" COMPOUND
WATER MIX POWDER
DENSITY - 1203.5 KG/M³
TEMPERATURE 0 °C TO 150 °C
BY INSTALLATION CONTRACTOR

REMOVE PLUG
CONNECT PURGE LINE

SECTION B-B
SCALE 1 : 4

SECTION C-C
SCALE 1 : 4

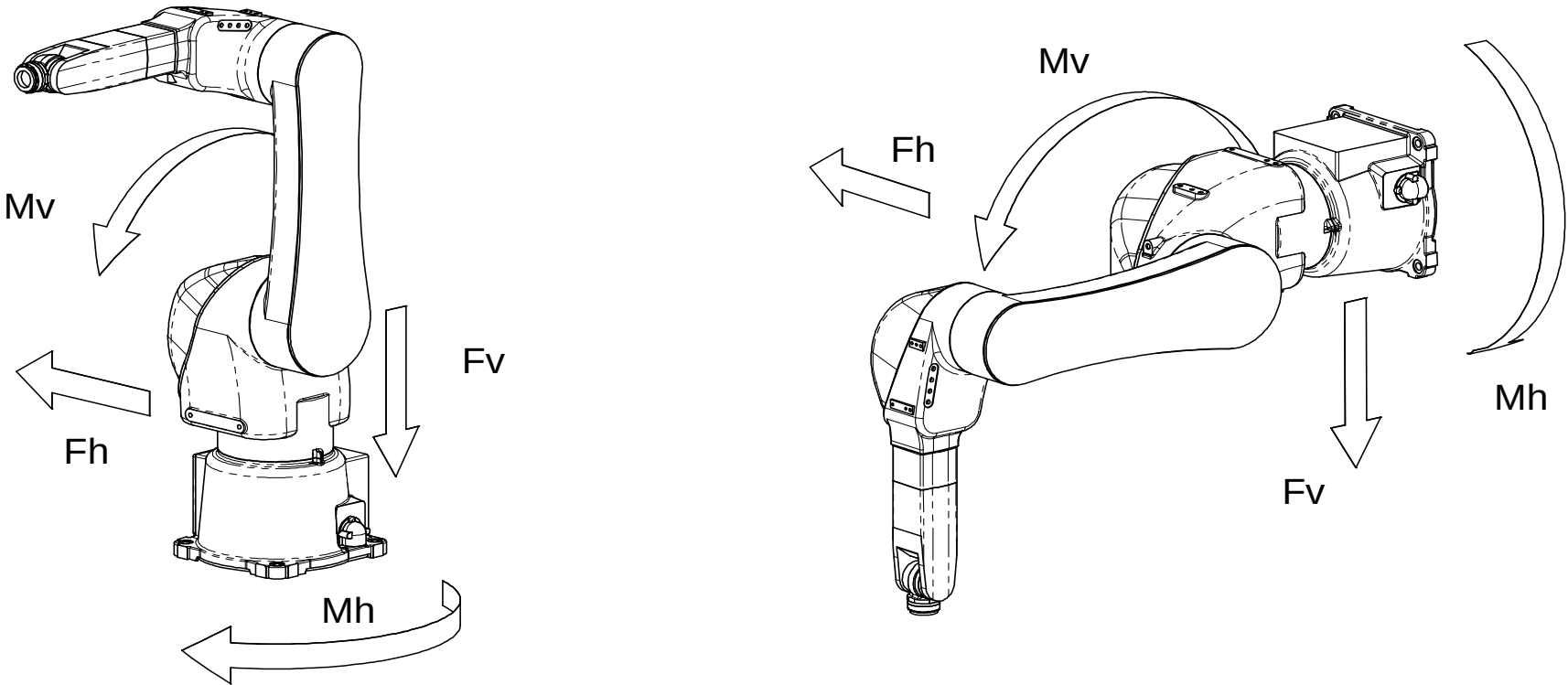
EO7347-134-000B
REQUIRED WHEN SEALOFF IS
MORE THAN 18" FROM BOOTH WALL

[ET-7347-840-001]
P-40iA CABLE PENETRATION

REV A SHEET 4 OF 4

4.2 Robot Mounting Interface and Loading

Figure 4-5 ET-7347-830-001 Sheet 1 of 2, P-40iA MOUNTING INTERFACE AND LOADING



WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

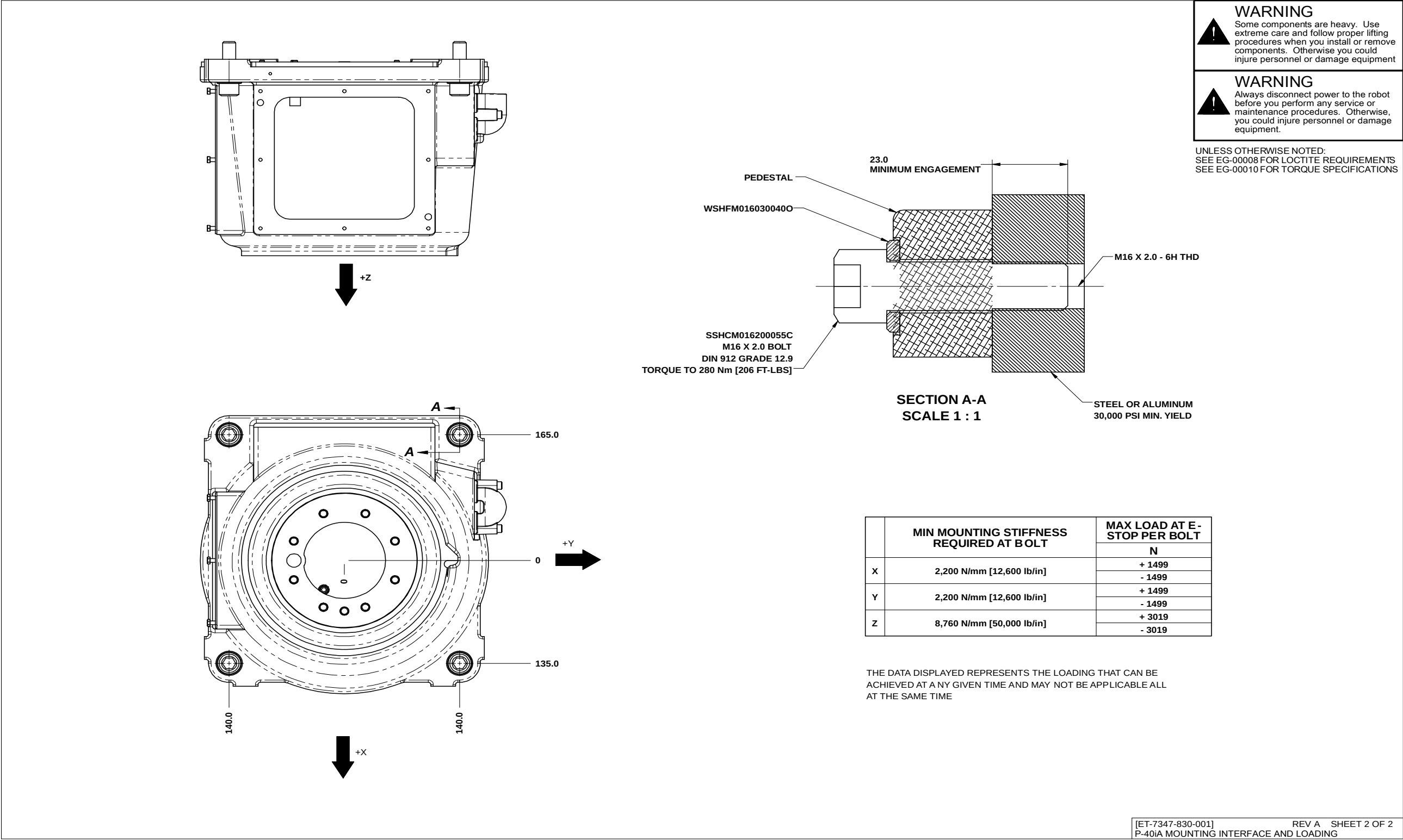
WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

	Floor Mount		Wall Mount	
	Dynamic	Estop	Dynamic	Estop
Fh	386 N	1265 N	334 N	774 N
Fv	1077 N	1843 N	1316 N	2025 N
Mh	369 Nm	1208 Nm	578 Nm	1433 Nm
Mv	948 Nm	1718 Nm	1332 Nm	2343 Nm

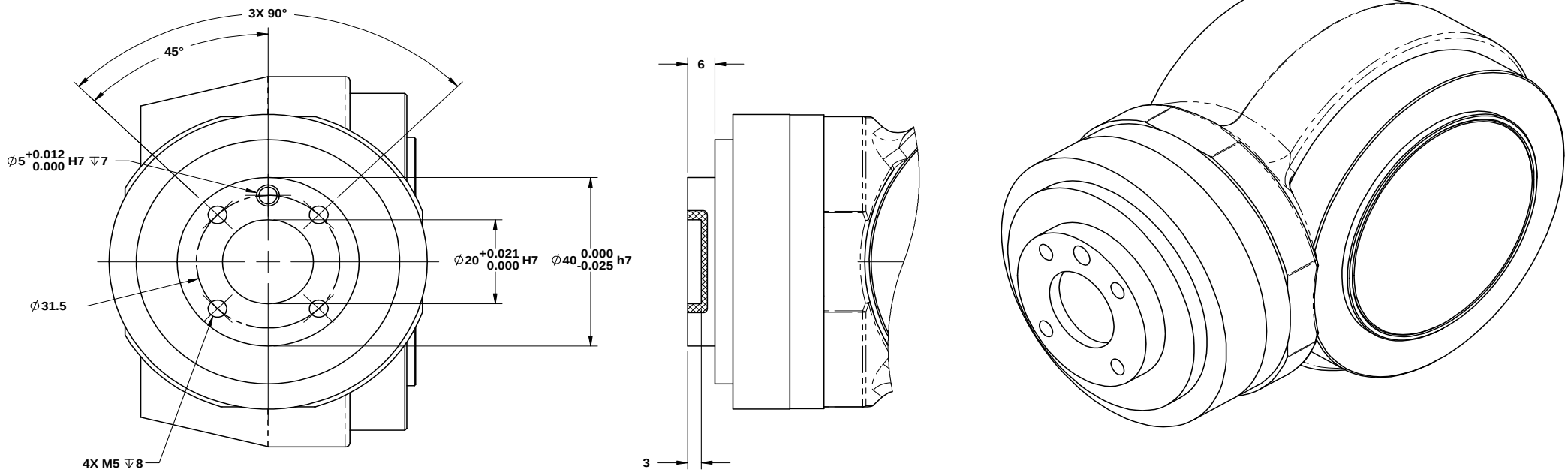
Standard Hardware Descriptions	
FANUC Part No.	Description
SSHCM016200055C	M16 X 2.0 X 55mm L.G. Steel Socket Head Cap
WSHFM016030040 O	M16 Flat Washer

Figure 4-6 ET-7347-830-001 Sheet 2 of 2, P-40iA MOUNTING INTERFACE AND LOADING



4.3 Wrist Interface and Payload Information

Figure 4-7 ET-7347-850-001 Sheet 1 of 3, P-40iA WRIST INTERFACE & PAYLOAD SPECIFICATIONS



WARNING

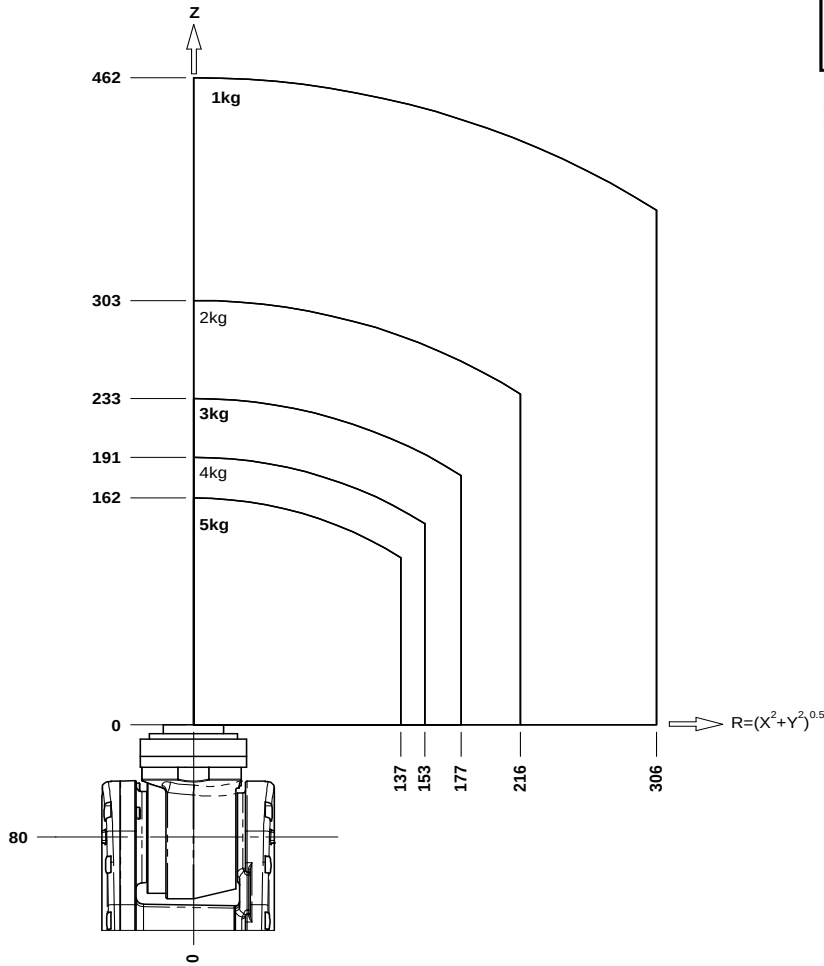
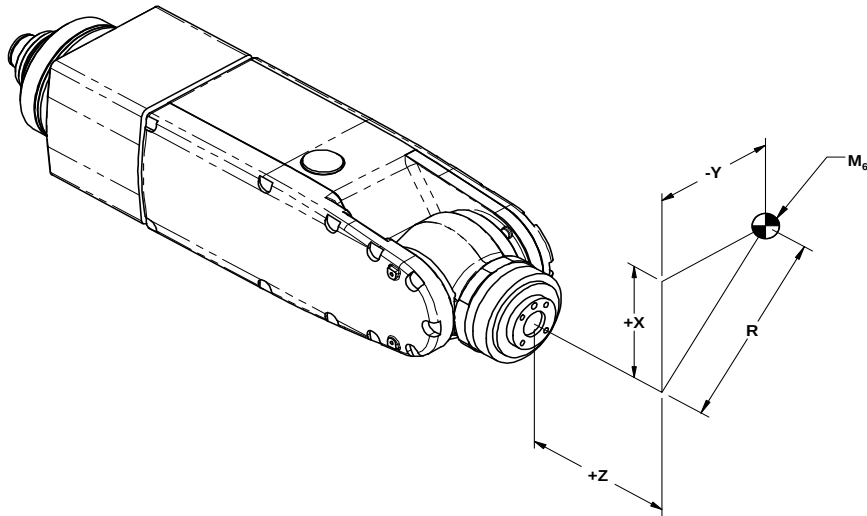
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING

Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-8 ET-7347-850-001 Sheet 2 of 3, P-40iA WRIST INTERFACE & PAYLOAD SPECIFICATIONS



ACCEPTABLE POINT MASS PAYLOAD CG LOCATIONS

MAX WRIST PAYLOAD (M ₆)	INERTIA			MOMENT		
	I _{J4}	I _{J5}	I _{J6}	M _{J4}	M _{J5}	M _{J6}
kg	kg m ²			Nm		
5	0.3	0.1		11.9	6.7	

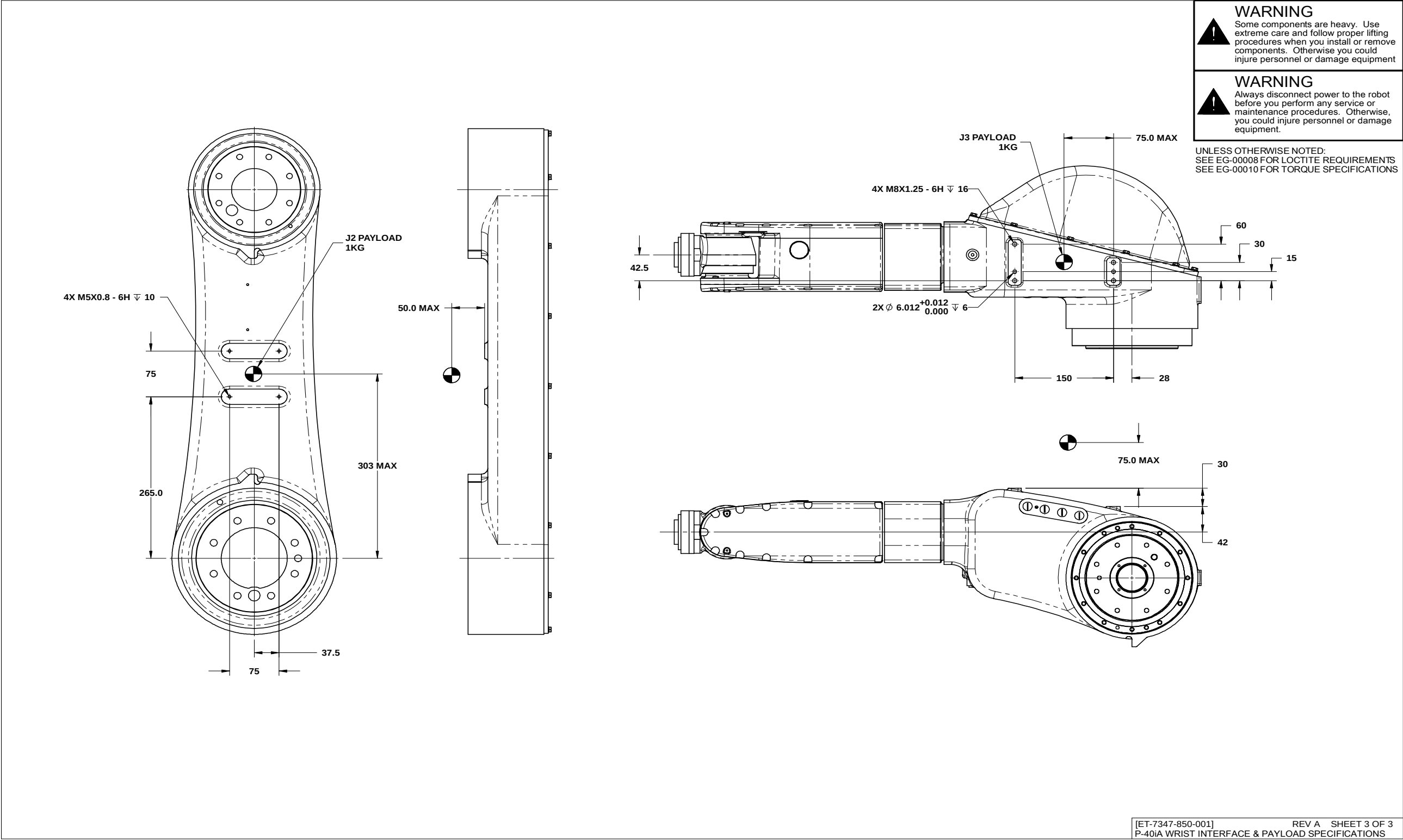
NOTES:
WRIST PAYLOAD INERTIAS AND CG LOCATIONS MUST BE SATISFIED
SIMULTANEOUSLY

WARNING
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

WARNING
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

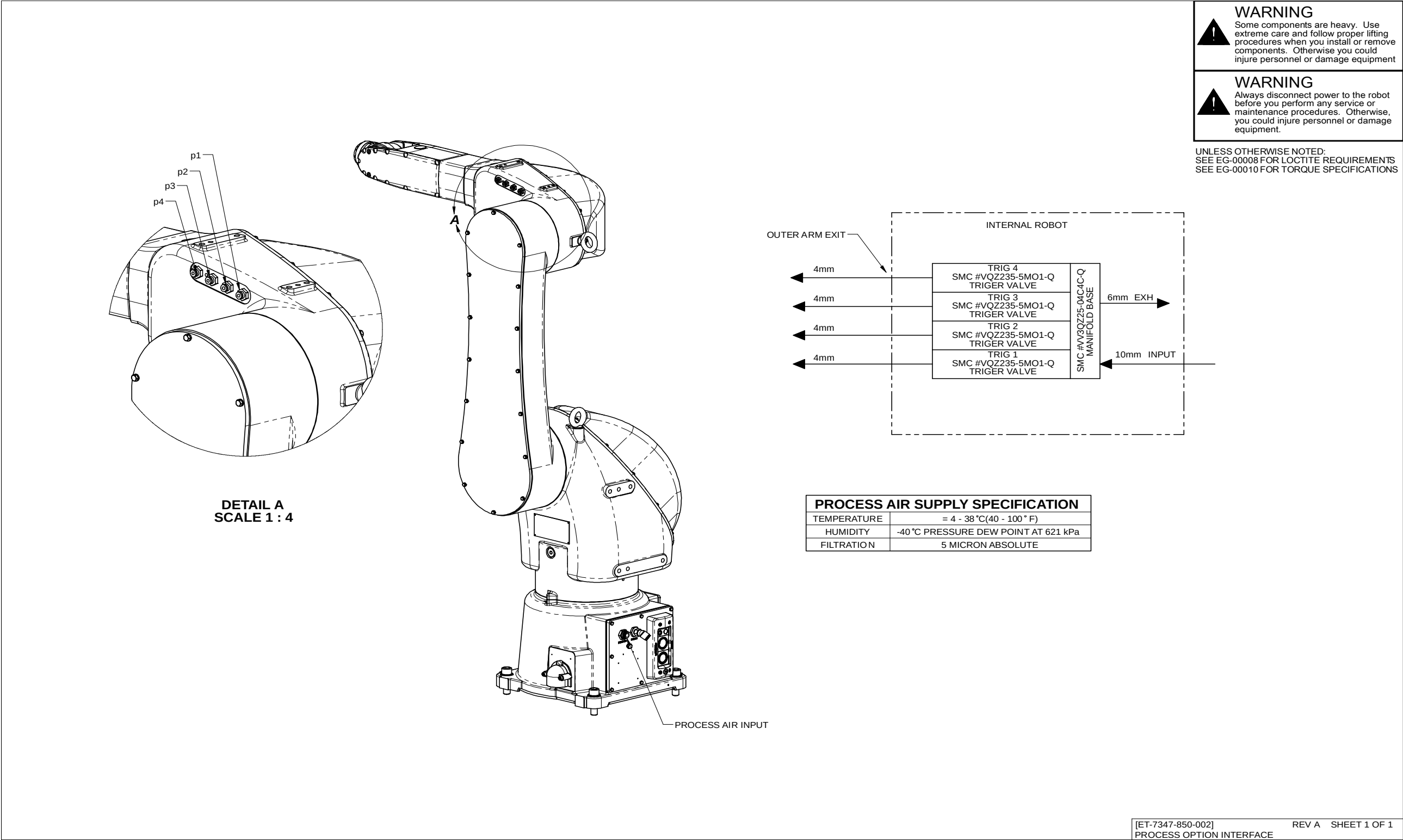
UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

Figure 4-9 ET-7347-850-001 Sheet 3 of 3, P-40iA WRIST INTERFACE & PAYLOAD SPECIFICATIONS



4.4 Process Option Interface

Figure 4-10 ET-7347-850-002, PROCESS OPTION INTERFACE

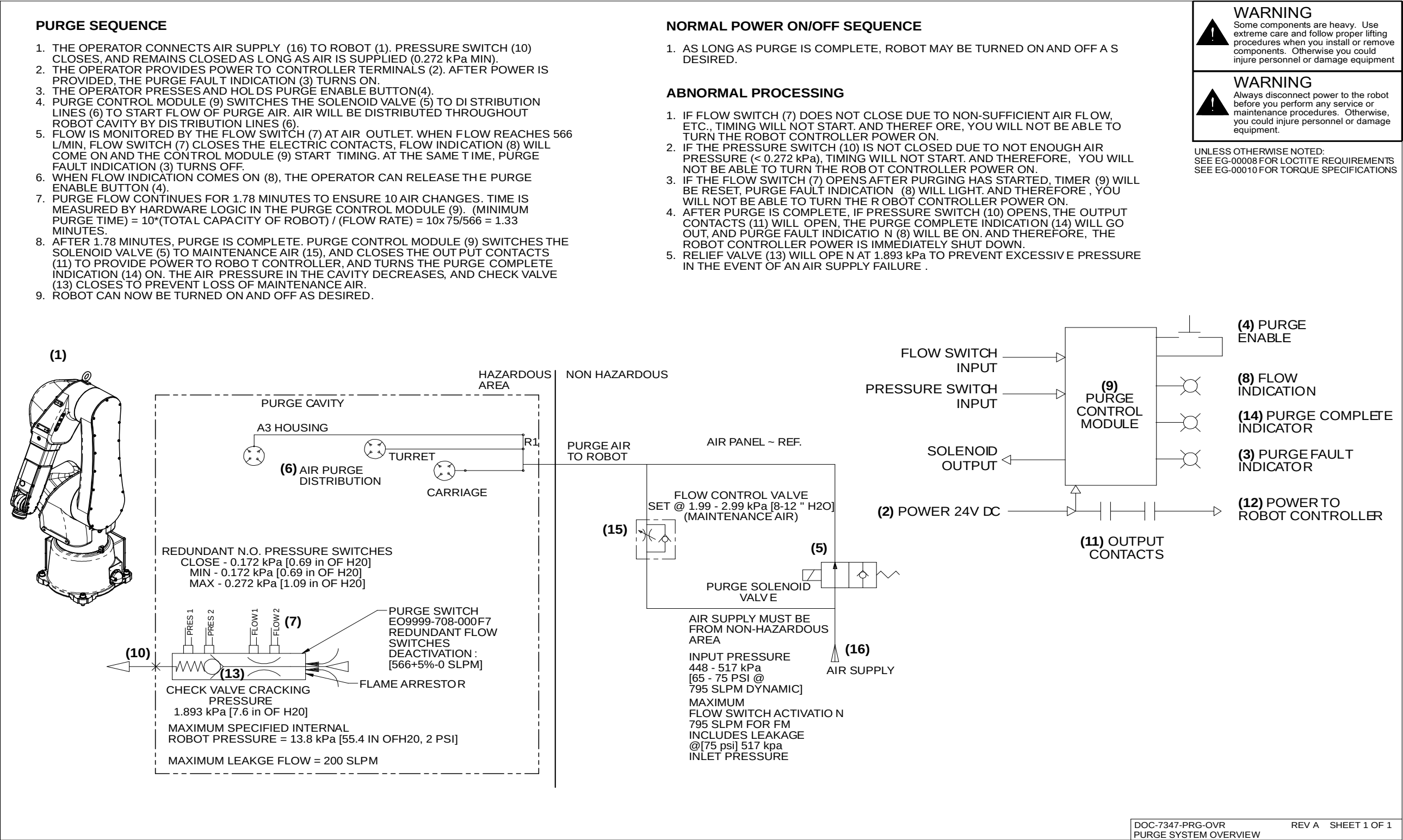


5

PURGE SYSTEM

5.1 Purge Overview

Figure 5-1 DOC-7347-PRG-OVR, PURGE SYSTEM OVERVIEW

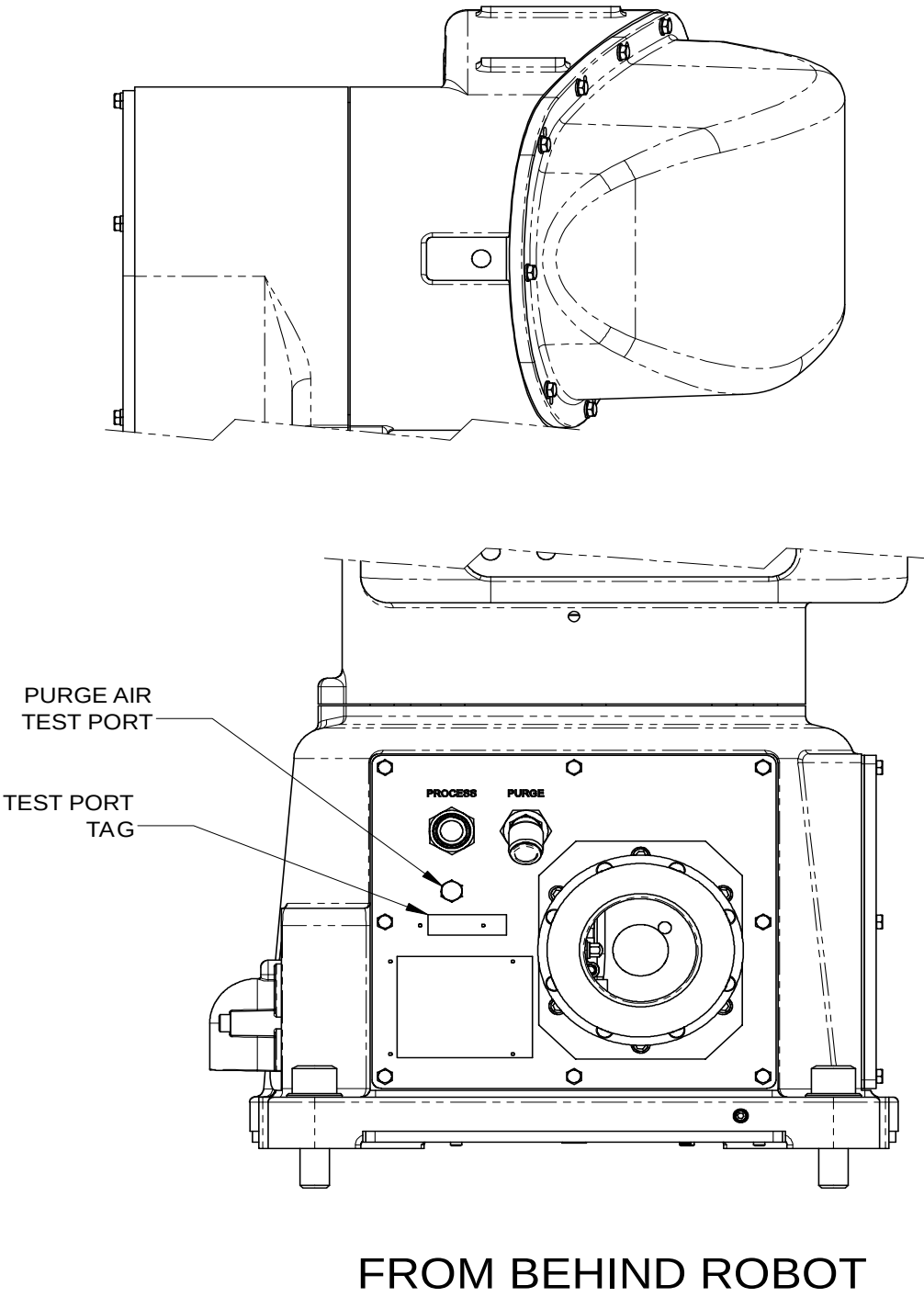


DOC-7347-PRG-OVR
PURGE SYSTEM OVERVIEW

REV A SHEET 1 OF 1

5.2 Purge Test Port Location

Figure 5-2 DOC-7347-PRG-TST, P-40iA PURGE TEST PORT LOCATION



**WARNING**
Some components are heavy. Use extreme care and follow proper lifting procedures when you install or remove components. Otherwise you could injure personnel or damage equipment.

**WARNING**
Always disconnect power to the robot before you perform any service or maintenance procedures. Otherwise, you could injure personnel or damage equipment.

UNLESS OTHERWISE NOTED:
SEE EG-00008 FOR LOCTITE REQUIREMENTS
SEE EG-00010 FOR TORQUE SPECIFICATIONS

6 MASTERING

6.1 Overview

When you master a robot you define the physical location of the robot by synchronizing the mechanical information with the robot’s positional information. A robot must be mastered to operate properly. Robots are usually mastered before they leave FANUC America Corporation. However, it is possible that a robot might lose its mastering data and need to be remastered.

The robot axes are controlled by a closed loop servo system. The controller outputs a command signal to drive each motor. A feedback device mounted on the motor, called a serial pulse coder, sends a signal back to the controller. During robot operation, the controller constantly analyzes the feedback signal, and modifies the command signal to maintain proper location and velocity of the end effector at all times.

For the robot to move accurately to recorded positions, the controller has to "know" the position of each axis. It does this by comparing the serial pulse coder reading during operation with a reading taken at a known mechanical reference point on the robot. Mastering records the serial pulse coder reading at a known mechanical reference point. This mastering data is stored with other user data in the controller memory and backed up by a lithium battery so that it is maintained when the controller is shut down.

When the controller is shut down under normal conditions, each serial pulse coder’s present reading is maintained in the pulse coder by backup batteries on the robot. (These batteries might be located in the controller or the robot auxiliary panel on P-series robots.) When the controller is turned on, the controller requests the stored reading from the serial pulse coder. When the controller receives the pulse coder reading, the servo system can operate normally. This process is called calibration. Calibration occurs automatically each time the controller is turned on.

When pulse coder backup battery power is interrupted while the controller is powered down, calibration fails at power up and the only robot motion possible is joint mode jogging. To restore proper operation, the robot must be mastered and calibrated.

**WARNING:**
When calibration has failed, the axis software travel limits will be ignored, allowing the robot to be jogged farther than normal. Be careful when jogging the robot in an uncalibrated condition or you could injure personnel or damage equipment.

**WARNING:**
If the robot has been configured with DCS (Dual Check Safety), consult the DCS manual for mastering instructions.

Note: The following procedures might be password-protected on your robot. If passwords have been set, you will need access at the Install level to perform mastering. Obtain the Install level password and log in at the Install level or you will be unable to perform the following procedures.

Before mastering the robot, you must clear any faults that prevent servo power from being restored or that prevent mastering completion.

Use Procedure 6-1 to clear common faults related to mastering and to prepare the robot for mastering. For more detailed information on fault recovery, refer to the *FANUC America Corporation Controller Maintenance Manual*.

If you are using a FANUC America Corporation A-series robot you should use a mastering fixture to master your robot. Fixture mastering is performed on **FANUC America Corporation P-series robots** by aligning mastering pins and surfaces on the robot. Refer to the *Mechanical Unit Service Manual* specific to your robot model for procedures on how to set up and use a mastering fixture.

If you are using a FANUC America Corporation M-series or S-series robot you can either master to a fixture or you can master to zero degrees. Refer to the *Mechanical Unit Service Manual* specific to your robot model for more

information about either of these methods.

If you are using a FANUC America Corporation P-series robot and you have zero degree witness marks scored onto your robot, then you can master the robot to zero degrees. Refer to the *Mechanical Unit Service Manual* specific to your robot model for procedures on how to master your robot.

Quick mastering is a convenient way to master a robot **after** you have recorded a reference position. You cannot quick master a robot unless the reference position was taught before mastering was lost.

**CAUTION:**
Record the quick master reference position after the robot is installed to preserve the factory mastering settings for future remastering.

Table 6-1 SYSTEM Master/Cal Items

ITEM	DESCRIPTION
SET QUICK MASTER REF	This item allows you to set master reference positions. You can save the master reference positions to the default device in case the robot has lost mastery due to an electrical or software problem.
FIXTURE POSITION MASTER	This item allows you to master the robot to a fixture. When you master to a fixture, you use a mastering fixture to align the robot axes and then record their serial pulse coder readings. You can master any robot to a fixture.
SINGLE AXIS MASTER	This item allows you to master a single axis of the robot. You can master a single axis of any robot provided that there is a reference mark at a known position on that axis. When a single axis of a robot is mastered, mastering data for the other axes remains unchanged.
QUICK MASTER	This item allows you to quick master the robot. Quick mastering allows you to minimize the time required to remaster the robot using a reference position you established when the robot was properly mastered. You cannot quick master the robot unless you have previously recorded this quick master reference position. Record the quick master reference position when the robot is properly mastered. The best time to record the quick master reference position is when the robot is still factory-mastered.
ZERO POSITION MASTER	This item allows you to master the robot to zero degrees. When you master to zero degrees, you position all axes at their zero degree witness marks and record their serial pulse coder readings. You can master any M-series or S-series robot to zero degrees. If you are using a FANUC America Corporation P-series robot, and you have witness marks scored onto your robot, you can master the robot to zero degrees.
CALIBRATE	This item allows you to calibrate the robot.
Message Field	This item is a status area at the bottom of the screen that lets you know when the robot has been successfully mastered or calibrated.

6.2 Resetting Alarms and Preparing For Mastering

When you turn on the robot after pulse coder backup battery power has been interrupted you will see a SRVO-062 BZAL alarm. You might also see a SRVO-038 Pulse Mismatch alarm. Before mastering the robot you must reset these alarms and rotate the motor of each axis that lost battery power to prepare the robot for mastering. Use Procedure 6-1 to reset these alarms and prepare the robot for mastering.

Procedure 6-1 Preparing the Robot for Mastering

Conditions

- You see a SRVO-062 BZAL or SRVO-038 Servo mismatch alarm.
- You are not mastering a P-200 robot.

Steps

- 1. If necessary, replace the robot batteries with four new 1.5 volt alkaline batteries, size D. Observe the direction arrows in the battery box for proper orientation of the batteries.
- 2. Press MENUS.
- 3. Select SYSTEM.
- 4. Press F1, [TYPE].
- 5. **If Master/Cal is not listed on the [TYPE] menu**, do the following; otherwise, continue to Step 6
 - a. Move the cursor to VARIABLE and press ENTER.
 - b. Move the cursor to \$MASTER_ENB.
 - c. Press the numeric key "1" and then press ENTER on the teach pendant.
 - d. Press F1, [TYPE].
- 6. Select Master/Cal. You will see a screen similar to the following.

**WARNING:**
For the M-6i (ARC Mate 100i), M-16i (ARC Mate 120i), and M-16iL (ARC Mate 120iL) robots, setting TORQUE to OFF using the F4, TORQUE function key on the SYSTEM Master/Cal screen releases the robot brakes. When the brakes are released, the robot arm will drop suddenly unless it is supported. DO NOT use this function key unless instructed to do so, otherwise, you could injure personnel or damage equipment.

```
SYSTEM Master/Cal
TORQUE = ON
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
```

Note: F4, TORQUE does not appear on the Master/Cal screen for all robot models. When F4, TORQUE appears on the Master/Cal screen, it allows the robot brakes to be released.

- 7. Press F3, RES_PCA. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Reset pulse coder alarm? [NO]
```

- 8. Press F4, YES. You will see a screen similar to the following.

```
SYSTEM Master/Cal
TORQUE = ON
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Pulse coder alarm reset!
```

Note: If you exit the Master/Cal screen by pressing F5, DONE, the Master/Cal screen will be hidden. Master/Cal will not be available by pressing F1, [TYPE]. To display the Master/Cal screen again, perform Step 1 through Step 6.

- 9. Turn off the controller.
- 10. Wait a few seconds, then turn the power disconnect circuit breaker on to turn the controller on again.
- 11. If the SRVO-062 BZAL alarm is still present; there is a battery, cable or pulsecoder problem. Refer to the *FANUC America Corporation Controller Maintenance Manual* for further information.
- 12. If a SRVO-038 Pulse Mismatch alarm is present at this time, repeat Step 1 through Step 8 to reset it. It is not necessary to restart the robot after resetting to clear this alarm.
- 13. If a SRVO-075 Pulse Not Established alarm is present at this time, press the RESET key to clear it.
- 14. Rotate each axis that lost battery power by at least one motor revolution in either direction. Failure to do so will result in the SRVO-075 Pulse Not Established alarm recurring and mastering will not be possible.
 - a. For each rotary axis, jog at least twenty degrees.
 - b. For each linear axis, jog at least thirty millimeters.
- 15. Perform any of the mastering procedures from the MASTER/CAL menu.

6.3 Saving and Restoring Mastering Data

You can save the master reference positions to the default device in case the robot has lost mastery due to an electrical or software problem. Use Procedure 6-2 to save and restore master reference position data.

Procedure 6-2 Saving and Restoring Master Reference Position Data

Conditions

- The robot is mastered. Refer to Section 6.4 through Section6.6 to choose a mastering method and master your robot.
- The default device is set.
- The robot is on and is working properly.

Steps

- 1. Press MENUS.
 - 2. Select SYSTEM.
 - 3. Press F1, [TYPE].
 - 4. Select Master/Cal.
- If Master/Cal is not listed on the [TYPE] menu,** do the following; otherwise, continue to Step 5
- a. Select VARIABLE from the [TYPE] menu.
 - b. Move the cursor to \$MASTER_ENB.
 - c. Press the numeric key "1" and then press ENTER on the teach pendant.
 - d. Press F1, [TYPE].
 - e. Select Master/Cal. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'Enter' or number key to select.
```

Note: F4, TORQUE does not appear on the Master/Cal screen for all robot models. When F4, TORQUE appears on the Master/Cal screen, it allows the robot brakes to be released.

- 5. Press FCTN.
- 6. Select SAVE. The file SYSMAST.SV is copied to the default device.
- 7. **To restore the saved file,** press F2, LOAD.

6.4 Single Axis Mastering

You can master a single axis of any robot provided that there is a reference mark at a known position on that axis. When a single axis of a robot is mastered, mastering data for the other axes remains unchanged. Use Procedure 6-3 to master a single axis.

Procedure 6-3 Mastering a Single Axis

Conditions

- You have cleared any servo faults that prevent you from jogging the robot.
- You have jogged each axis that has lost mastering at least one motor turn. See Procedure 6-1

Steps

- 1. Jog the unmastered axis of the robot to the single axis mastering position.
 - **For M-series or S-series robots only:** Using the joint coordinate system, jog the unmastered axis of the robot to the zero degree witness mark. If you are single axis mastering the J2-axis, the J3-axis must first be lined up at its zero degree mark. Otherwise the positions of the other axes are unimportant. Refer to the *Mechanical Service Manual* specific to your robot model for the location of the zero degree witness marks.
 - **For P-series robots only:** Using the joint coordinate system, jog the unmastered axis of the robot to the standard mastering location and align the mark, pin or surface using a mastering block or straight edge as required. If you are single axis mastering the J3-axis, the J2-axis must first be lined up at its standard mastering location. Otherwise, the positions of the other axes are not important.
- 2. Press MENUS.
- 3. Select SYSTEM.
- 4. Press F1, [TYPE].
- 5. **If Master/Cal is not listed on the [TYPE] menu,** do the following; otherwise, continue to Step 6
 - a. Select VARIABLE.
 - b. Move the cursor to \$MASTER_ENB.
 - c. Press the numeric key "1" and then press ENTER on the teach pendant.
 - d. Press F1, [TYPE].
- 6. Select Master/Cal. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
```

Note: F4, TORQUE does not appear on the Master/Cal screen for all robot models. When F4, TORQUE appears on the Master/Cal screen, it allows the robot brakes to be released.

7. Move the cursor to SINGLE AXIS MASTER and press ENTER. You will see a screen similar to the following

SINGLE AXIS MASTER				
ACTUAL POS	(MSTR POS)	(SEL)	[ST]	
J1 0.000	(0.000)	(0)	[2]	
J2 3.514	(35.000)	(0)	[0]	
J3 -7.164	(-100.000)	(0)	[2]	
J4 -35.366	(0.000)	(0)	[2]	
J5 -1.275	(-80.000)	(0)	[2]	
J6 4.571	(0.000)	(0)	[2]	
E1 0.000	(0.000)	(0)	[0]	
E2 0.000	(0.000)	(0)	[0]	
E3 0.000	(0.000)	(0)	[0]	

Note: A "0" in the [ST] column indicates that the axis is unmastered.

8. Move the cursor to the (MSTR POS) column and move it up or down to the unmastered axis. (Any unmastered axis will have the number 0 in the [ST] column).
9. **For M-series or S-series robots only:** Enter the position where single axis mastering is to be performed in the (MSTR POS) column for the unmastered axis.
- a. For robots where single axis mastering is performed at the zero degree position, press the numeric key "0" and press ENTER.
 - b. For robots where single axis mastering is performed at the fixture position, enter the fixture position and press ENTER.

Note: Some P-series robots are single axis mastered at the fixture position. All other robots are single axis mastered at the zero degree position.

10. Continuously press and hold the DEADMAN switch and turn the teach pendant ON/OFF switch to ON.
11. Press F4, GROUP. Enter the appropriate group number. Enter 1 for a left robot or 2 for a right robot. The left and right designation is determined by looking downstream of the conveyer travel.
12. Move the cursor to the SEL column and move it up or down to the unmastered axis.
13. Press the numeric key "1" and press ENTER.
14. Press F5, EXEC. Mastering will be performed automatically. You will see a screen similar to the following.

SINGLE AXIS MASTER				
ACTUAL POS	(MSTR POS)	(SEL)	[ST]	
J1 0.000	(0.000)	(0)	[2]	
J2 3.514	(0.000)	(0)	[2]	
J3 -7.164	(-100.000)	(0)	[2]	
J4 -35.366	(0.000)	(0)	[2]	
J5 -1.275	(-80.000)	(0)	[2]	
J6 4.571	(0.000)	(0)	[2]	
E1 0.000	(0.000)	(0)	[0]	
E2 0.000	(0.000)	(0)	[0]	
E3 0.000	(0.000)	(0)	[0]	

Note: The J2 axis is now mastered and a 2 is displayed in the [ST] column.

- 15. Press PREV.
- 16. Move the cursor to CALIBRATE and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Calibrate ? [NO]
```

- 17. Press F4, YES. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Robot Calibrated! Cur Jnt Ang(deg):
<0.000> <0.000> <-7.164>
<-35.366> <-1.275> <4.571>
```

Note: If you exit the Master/Cal screen by pressing F5, DONE, the Master/Cal screen will be hidden. Master/Cal will not be available by pressing 1, [TYPE]. To display the Master/Cal screen again perform Step 1 through Step 6

6.5 Quick Mastering

Quick mastering allows you to minimize the time required to remaster the robot using a reference position you established when the robot was properly mastered. You cannot quick master the robot unless you have previously recorded this quick master reference position.

Record the quick master reference position when the robot is properly mastered. The best time to record the quick master reference position is when the robot is still factory-mastered.

If calibration fails because pulse coder backup battery power has been interrupted, you can use this reference position to master the robot in a minimum amount of time. When the zero position marks do not line up because of mechanical disassembly or repair, you must master to a fixture or perform zero degree mastering.

You can define a quick master reference position and perform quick mastering on any robot model.

Use Procedure 6-4 to record the quick master reference position. Use Procedure 6-5 to quick master the robot.

**CAUTION:**
Record the quick master reference position after the robot is installed to preserve the factory mastering settings for future remastering.

Procedure 6-4 Recording the Quick Master Reference Position

Conditions

- The robot is properly mastered.

Steps

1. Align each axis of the robot with the reference mark you have chosen as a quick master reference position.

Note: It is convenient to use the zero degree marks for the quick master reference position Refer to the *Mechanical Unit Service Manual* specific to your robot model for the location of the zero degree marks. But if you prefer, you can use any robot position as long as you scribe witness marks on each axis at the reference point.

2. Press MENUS.
3. Select SYSTEM.
4. Press F1, [TYPE].
5. **If Master/Cal is not listed on the [TYPE] menu**, do the following; otherwise, continue to Step 6
 - a. Select VARIABLE.
 - b. Move the cursor to \$MASTER_ENB.
 - c. Press the numeric key "1" and then press ENTER on the teach pendant.
 - d. Press F1, [TYPE].
6. Select Master/Cal. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
```

Note: F4, TORQUE does not appear on the Master/Cal screen for all robot models. When F4, TORQUE appears on the Master/Cal screen, it allows the robot brakes to be released.

7. Move the cursor to SET QUICK MASTER REF and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Set quick master reference ? [NO]
```

8. Press F4, YES. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Quick Master Reference Set!
```

Note: If you exit the Master/Cal screen by pressing F5, DONE, the Master/Cal screen will be hidden. Master/Cal will not be available by pressing F1, [TYPE]. To display the Master/Cal screen again, perform Step 1 through Step 6

Procedure 6-5 Quick Mastering the Robot

Conditions

- Calibration has failed because backup battery power has been interrupted.

Note If the zero degree marks do not line up because of mechanical disassembly or repair, you cannot perform this procedure. In this case, master to a fixture or master to zero degrees to restore robot mastering.

- The quick master reference position was recorded before calibration failed.
- You have cleared any servo faults that prevent you from jogging the robot.
- You have jogged each axis that has lost mastering at least one motor turn. See Procedure 6-1

Steps

1. Jog the robot to the quick master reference position that has previously been recorded.
2. Press MENUS.
3. Select SYSTEM.
4. Press F1, [TYPE].
5. **If Master/Cal is not listed on the [TYPE] menu**, do the following; otherwise, continue to Step 6
 - a. Select VARIABLE.
 - b. Move the cursor to \$MASTER_ENB.
 - c. Press the numeric key "1" and then press ENTER on the teach pendant.
 - d. Press F1, [TYPE].
6. Move the cursor to Master/Cal and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
```

Note: F4, TORQUE does not appear on the Master/Cal screen for all robot models. When F4, TORQUE appears on the Master/Cal screen, it allows the robot brakes to be released.

7. Move the cursor to QUICK MASTER and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Quick master ? [NO]
```

8. Press F4, YES. Mastering will be performed automatically. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Robot Mastered! Mastering Data:
<0> <11808249> <38767856>
<9873638> <122000309> <2000319>
```

9. Move the cursor to CALIBRATE and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Calibrate ? [NO]
```

10. Press F4, YES. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Robot Calibrated! Cur Jnt Ang(deg):
<0.003> <-0.0012> <-0.050>
<0.009> <0.015> <-0.002>
```

Note: If you exit the Master/Cal screen by pressing F5, DONE, the Master/Cal screen will be hidden. Master/Cal will not be available by pressing F1, [TYPE]. To display the Master/Cal screen again, perform Step 1 through Step 6

6.6 Mastering To A Fixture (Fixture Position Master)

When you master to a fixture, you use a mastering fixture to align the robot axes and then record their serial pulse coder readings. You can master any robot to a fixture. P-series paint robots do not use fixtures. A mastering block, and in some cases, a steel straight edge or ruler is used to align mastering surfaces on P-series paint robots. A mastering block is used for aligning the mastering surfaces of the modular in-booth rail.

Use Procedure 6-6 to master to a fixture. **Note** that not all steps refer to paint robots.

Procedure 6-6 Mastering to a Fixture

Conditions

- You have the appropriate mastering fixture for your robot. (P-series robots require the robot to be put in a fixture position only. They do not actually use a fixture.)
- You have cleared any servo faults that prevent you from jogging the robot.
- You have jogged each axis that has lost mastering at least one motor turn. See Procedure 6-1

Steps

1. If automatic brake control is enabled, disable it as follows; otherwise continue to Step 2. This step is not for paint robots.
 - a. Press MENUS.
 - b. Select SYSTEM.
 - c. Press F1, [TYPE].
 - d. Select VARIABLE.
 - e. Move the cursor to \$PARAM_GROUP and press ENTER twice.
 - f. Move the cursor to \$SV_OFF_ALL and press F5, FALSE.
 - g. Move the cursor to \$SV_OFF_ENB and press ENTER.
 - h. Move the cursor to each line where the value is TRUE, and press F5, FALSE.
 - i. Turn off the controller.
 - j. Turn on the controller.
2. Install the mastering fixture on the robot and jog the robot into the mastering position. Refer to the *Mechanical Service Manual* specific to your robot model for the procedures for setup and use of a mastering fixture. For P-series robots align the mastering surfaces using the appropriate mastering block or straight edge. The modular in-booth rail uses a block.
3. Press MENUS.
4. Select SYSTEM.
5. Press F1, [TYPE].
6. **If Master/Cal is not listed on the [TYPE] menu**, do the following; otherwise, continue to Step 7
 - a. Move the cursor to VARIABLE and press ENTER.
 - b. Move the cursor to \$MASTER_ENB.
 - c. Press the numeric key "1" and then press ENTER on the teach pendant.
 - d. Press F1, [TYPE].
7. Select Master/Cal. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
```

Note: F4, TORQUE does not appear on the Master/Cal screen for all robot models. When F4, TORQUE appears on the Master/Cal screen, it allows the robot brakes to be released. This step is not for paint robots.

8. Move the cursor to FIXTURE POSITION MASTER and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Master at master position ? [NO]
```

9. Press F4, YES. Mastering will be performed automatically. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Robot Mastered! Mastering Data:
<0> <11808249> <38767856>
<9873638> <122000309> <2000319>
```

10. Move the cursor to CALIBRATE and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Calibrate ? [NO]
```

11. Press F4, YES. You will see a screen similar to the following

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Robot Calibrated! Cur Jnt Ang(deg):
<0.000> <29.226> <-100.625>
<0.000> <-79.375> <0.000>
```

Note: If you exit the Master/Cal screen by pressing F5, DONE, the Master/Cal screen will be hidden. Master/Cal will not appear on the SYSTEM F1, [TYPE] submenu. To see the Master/Cal screen again perform Step 1 through Step 7. Steps 12 —15 are not for paint robots.

12. For all robots except paint robots, move the wrist clear of the mastering fixture and remove the mastering fixture from the robot. Refer to the *Mechanical Service Manual* or *Mechanical Connection and Maintenance Manual* specific to your robot model on how to use a mastering fixture.
13. If you have disabled automatic brake control (this step is not for paint robots) in Step 1 , enable it as follows:
- a. Press MENUS.
 - b. Select SYSTEM.

- c. Press F1, [TYPE]
- d. Select VARIABLE.
- e. Move the cursor to \$PARAM_GROUP and press ENTER twice.
- f. Move the cursor to \$SV_OFF_ALL and press F4, TRUE.
- g. Move the cursor to \$SV_OFF_ENB and press ENTER.
- h. Move the cursor to each line where you previously changed the value, and press F4, TRUE.

14. Turn off the controller.
15. Wait a few seconds, then press the ON/OFF button on the operator panel to turn the controller on again.

6.7 Zero Degree Mastering

When you master to zero degrees, you position all axes at their zero degree witness marks and record their serial pulse coder readings. You can master any M-series or S-series robot to zero degrees. If you are using a FANUC America Corporation P-series robot, and you have witness marks scored onto your robot, you can master the robot to zero degrees. Use Procedure 6-7 to master to zero degrees

Procedure 6-7 Mastering to Zero Degrees

Conditions

- You have cleared any servo faults that prevent you from jogging the robot.
- You have jogged each axis that has lost mastering at least one motor turn. Refer to Procedure 6-1

Steps

1. Using the joint coordinate system, jog each axis of the robot to the zero degree witness mark. Refer to the *Mechanical Service Manual* specific to your robot model for the location of the witness marks.
2. Press MENUS.
3. Select SYSTEM.
4. Press F1, [TYPE].
5. **If Master/Cal is not listed on the [TYPE] menu**, do the following; otherwise, continue to Step 6
 - a. Select VARIABLE.
 - b. Move the cursor to \$MASTER_ENB.
 - c. Press the numeric key "1" and then press ENTER on the teach pendant.
 - d. Press F1, [TYPE].
6. Select Master/Cal. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
```

Note: F4, TORQUE does not appear on the Master/Cal screen for all robot models. When F4, TORQUE appears on the Master/Cal screen, it allows the robot brakes to be released.

7. Move the cursor to ZERO POSITION MASTER and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Master at zero position ? [N0]
```

8. Press F4, YES. Mastering will be performed automatically. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Robot Mastered! Mastering Data:
<0> <11808249> <38767856>
<9873638> <122000309> <2000319>
```

9. Move the cursor to CALIBRATE and press ENTER. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Press 'ENTER' or number key to select.
Calibrate ? [N0]
```

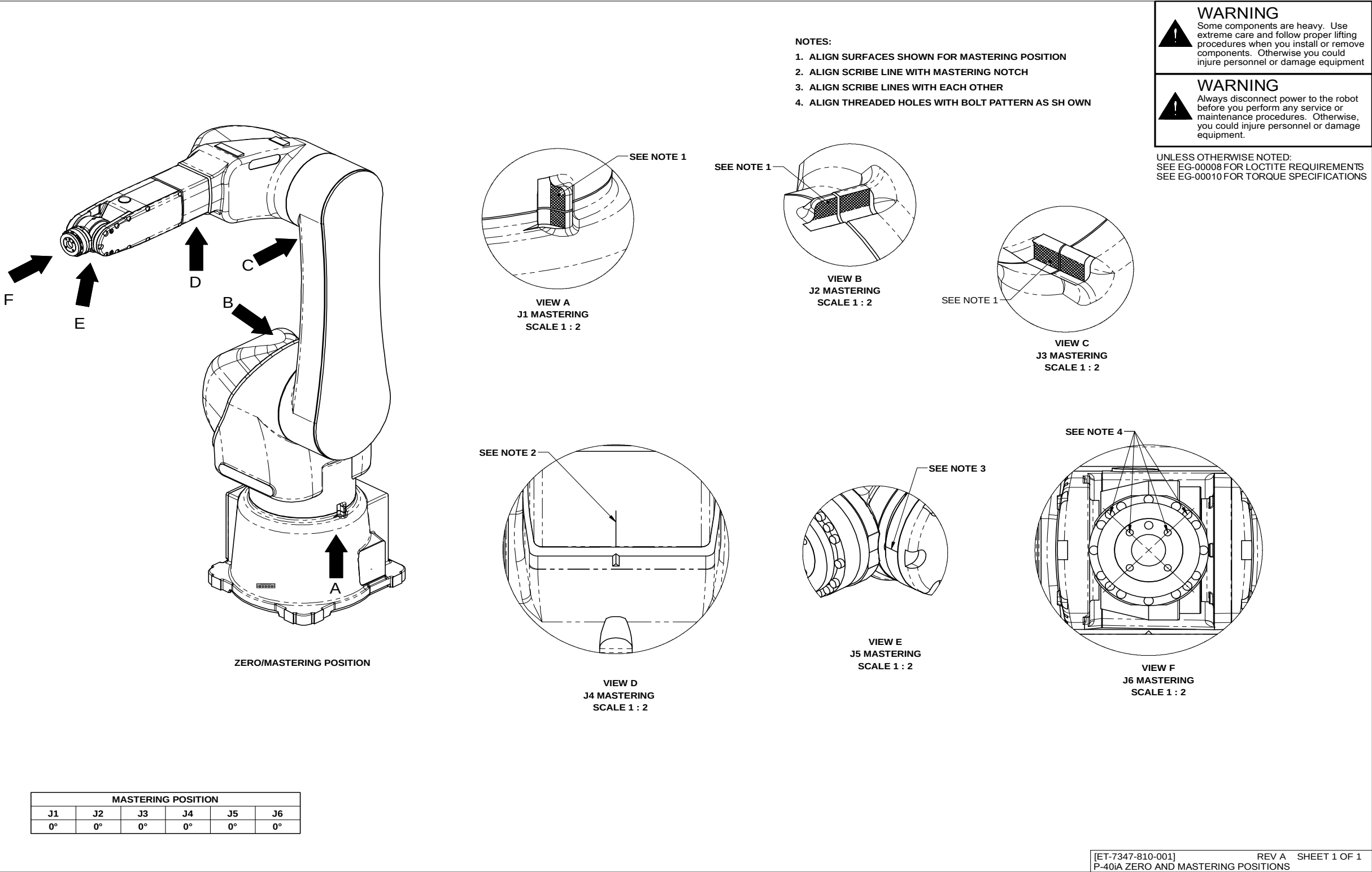
10. Press F4, YES. You will see a screen similar to the following.

```
SYSTEM Master/Cal
1 FIXTURE POSITION MASTER
2 ZERO POSITION MASTER
3 QUICK MASTER
4 SINGLE AXIS MASTER
5 SET QUICK MASTER REF
6 CALIBRATE
Robot Calibrated! Cur Jnt Ang(deg):
<0.000> <0.000> <0.000>
<0.000> <0.000> <0.000>
```

Note: If you exit the Master/Cal screen by pressing F5, DONE, the Master/Cal screen will be hidden. Master/Cal will not be available by pressing F1, [TYPE]. To display the Master/Cal screen again perform Step 1 through Step 6

6.8 Mastering Locations

Figure 6-1 ET-7347-810-001, P-40iA ZERO AND MASTERING POSITIONS



7 TROUBLESHOOTING

7.1 Mechanical Unit Troubleshooting

Table 7-1 shows the most common robot mechanical unit problems and their causes. If a cause or remedy is unclear, contact FANUC America Corporation.

Table 7-1 Problems and Causes

Symptom	Cause	Remedy
The robot does not return to the zero position correctly	Low voltage of the memory backup battery.	Replace the battery.
	Unusual external start signal.	Investigate the cause.
Position offset	The robot has been hit.	Revise the teaching points.
	Incorrect Mastering	Remaster Robot
	The robot is not firmly fastened to the base plate.	Retighten.
	Peripheral device was shifted.	Reposition.
	Defective pulse coder cable.	Replace the cable.
	APC (Absolute Pulse Coder) malfunction.	Replace the motor or the pulse coder.
	Play in mechanical unit.	Refer to excessive play symptom in this table.
Vibration	The robot is not firmly fastened to the base.	Retighten.
	Excessive floor vibration (especially when installed above the first floor).	Investigate the installation location and adjust the robot location if necessary.
	An incorrect servo adjustment.	Adjust the servo.
	A defective cable.	Replace the cable.
	The robot is not grounded.	Connect to ground.
	A defective motor.	Replace the motor.
	A defective reducer.	Replace the reducer.
Excessive play	Play in the mechanical unit.	Refer to the excessive play symptom in this table.
	A loose screw or pin.	Tighten and apply LOCTITE as specified.
	A defective reducer.	Replace the reducer.
	Worn gears.	Replace the gears.
	Worn bearings.	Replace the bearings.
Unusual noise	A cracked casting.	Replace cracked parts.
	Insufficient gear or reducer lubrications.	Lubricate.
	Foreign matter in a gear or a reducer.	Clean and lubricate.
	Excessive pressure on the bearings.	Reduce the pressure on the bearings.
	A defective reducer.	Replace the reducer.
	An incorrect gear adjustment.	Adjust the gears.
	Worn gears.	Replace the gears.
	Worn bearings.	Replace the bearings.
Over heating	Cable pair damage.	Replace the cable pair.
	Insufficient gear or reducer lubrication.	Lubricate.
	The specified grease was not used.	Replace with the specified grease.
	Excessive pressure on bearings.	Reduce the pressure on the bearings.
	Insufficient maintenance air	Increase maintenance air supply
The robot drops (10 mm or more) when the power supply is off	An excessive load on the wrist.	Reduce the load and restrict the movement conditions.
	The brake gap is too large.	Replace the motor.
Grease leakage	Fused brake drive relay.	Replace the relay.
	Deteriorated O-ring, oil seal, or packing.	Replace the O-ring, oil seal, or packing.
	A cracked casting.	Replace the cracked parts.
	Loose screws.	Tighten the screws.
	Excessive quantity of grease.	Remove some of the grease.

7.2 Purge Cycle Troubleshooting, Diagnostics, and Testing

7.2.1 Purge Procedure and Troubleshooting

Figure 7-1 PURGE PROCESS AND TROUBLESHOOTING FLOW CHART

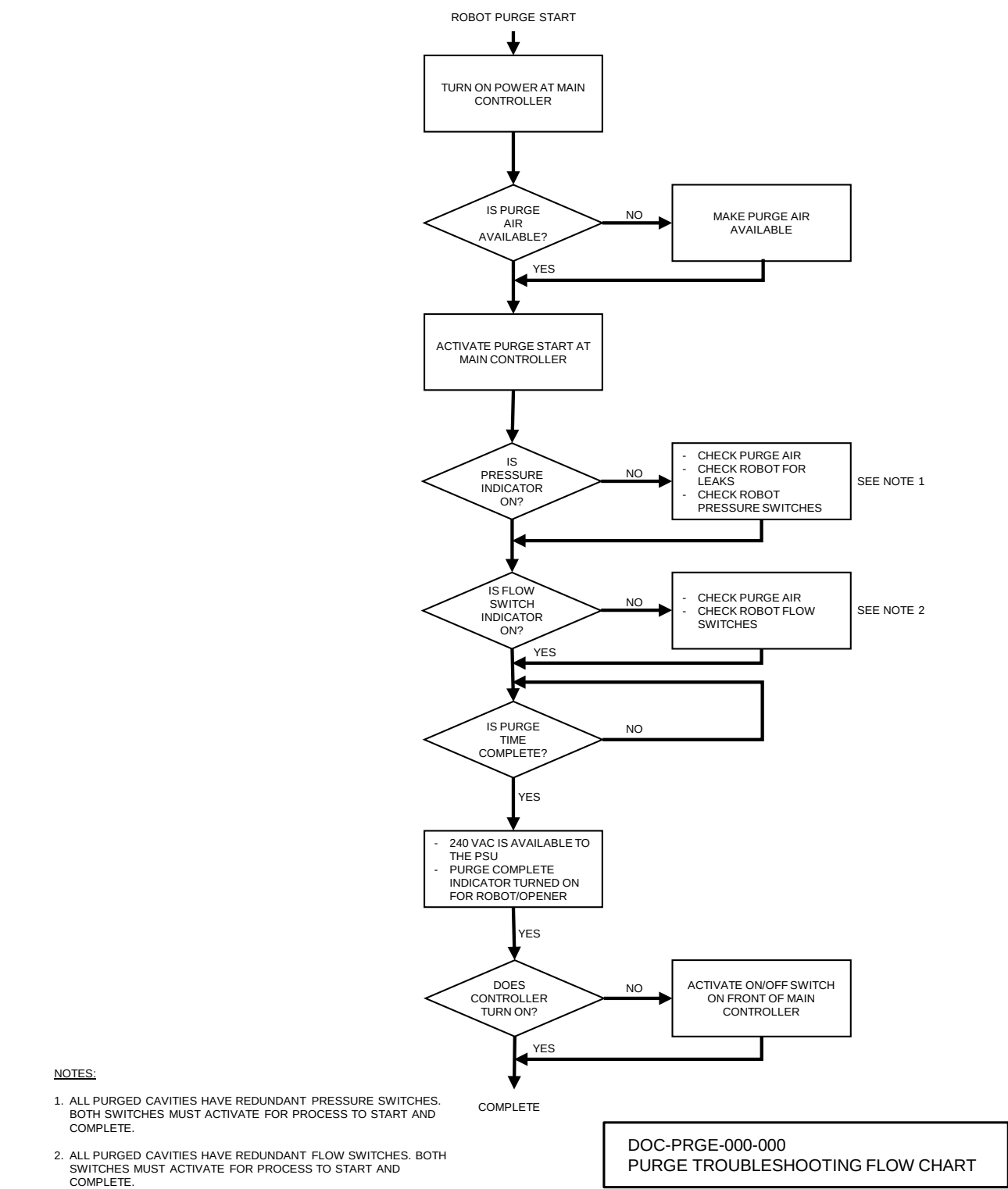


Table 7-2 Possible Purge Cycle Problems

PROBLEM	CAUSE	WHAT TO CHECK FOR
Inadequate air flow through the robot or opener	Air supply capacity inadequate	The air supply is inadequate. Check the air pressure at the pneumatic air supply.
Air leakage from robot or opener	Bad seal	Look for leakage around cover gaskets/bearing seals, at the cable entrance to the robot, holes where plugs or screws are missing, and through unsealed conduit.
Pressure switch or air flow switch does not appear to work	Defective pressure switch or air flow switch	Perform the pressure switch test and air flow switch test to check components.
Inconsistent readings on the pressure gauge	Air leaks or other problems with the pneumatic air distribution	Check continuity of the pressure switch and air flow switch, during dynamic test conditions. Check pressure gauge.

**CAUTION:**
Both the Pressure Switches and the Air Flow Switches are adjusted and sealed at the factory or authorized vendor. They are not adjustable in the field. If you break or tamper with the seal you will void FM certification on the robot.

7.2.2 Purge System Testing and Calibration

The following hardware listed below is necessary to perform the purge cycle test. Use Table 7-2 to test for purge air leakage.

Required Purge Hardware:

- Air source capable of supplying a minimum of 1100 slpm.
- One pressure gauge to read 0-15 inches of water (Dwyer #2015 or equivalent)
- Two air fittings, 1/8 inch NPT to 1/4 inch tubing (Parker #68C4-2 or equivalent)
- One nylon tube measuring 1/4 inch in diameter by 24 inches long (Parker #NN-4-035 or equivalent)
- Leak detector liquid (Nupro Snoop or equivalent)

Procedure 7-1 Testing for Purge Leaks

Steps

1. While the purge air is flowing, inspect the robot for leakage around all joints, seals and gaskets. No excessive air must be felt except at the purge outlet.

Note: Apply a leak detection fluid, such as Nupro Snoop, to areas under inspection; check for bubbling.

2. Pay close attention to:
 - Covers - Make sure each bolt is snug. Check for leaks around bolts and from cover, gasket, or surface damage.
 - Seals - Check each joint for leaks. Internal seals might be damaged or missing and internal mating surfaces, which also act as seals, might be worn or damaged.
 - Cable Penetration (Rox for CE and TIIS only) - Ensure that there is a tight seal around the cables.

Procedure 7-2 Testing the Purge Cycle for the Robot

- Make sure all robot covers, plugs, and connectors at the cable entrance to the robot and rail are tight.
- Ensure that the main disconnect on the controller is ON.
- Ensure that air is supplied to the system.

Steps

1. Ensure that the system air supply valve is open.
2. Tighten cover bolts where air leaks. Replace the gasket if necessary.
3. Connect a pressure gauge (0 to 15 inches H₂O) to the test port on the robot, see section **5.2 Purge Test Port Location**. Verify that the internal relative pressure is 8 - 10 inches H₂O. If the pressure is not at this level, verify that the regulator pressure is at 85 psi dynamic.
4. Be sure that the Purge Air Pressure indicator on the Main controller is illuminated.
5. If the Purge Air Pressure is not illuminated, refer to Table 7-2.
6. Press and hold the PURGE START button on the controller until it lights. After it is lit, the robot will purge for 5 minutes. If the purge enable indicator does not light shortly after the purge button is pressed (15 to 20 seconds), or if the light flickers, an air leak in the robot or an inadequate air source is the probable cause.

**WARNING:**
Connections to the pressure switches, flow switches, and solenoid valve are intrinsically safe. Ensure at least 50mm of clearance between intrinsically safe wiring and any other wiring. Ensure that any test equipment used in connection to intrinsically safe circuits is rated for use in a hazardous environment and has been certified by a nationally recognized testing laboratory such as UL, FM, CSA, and so forth. Otherwise, if you do not observe these precautions, it could cause an explosion, potentially resulting in injured personnel and damaged equipment.

8 ELECTRICAL

The drawings in this section are for reference only. See FANUC America Corporation *R-30iB Mate Plus Controller Maintenance* for detailed controller and robot electrical content.

NON-HAZARDOUS

HAZARDOUS (CLASSIFIED)

BOOTH WALL

NON-HAZARDOUS

HAZARDOUS (CLASSIFIED)

BOOTH WALL

ROBOT PURGED CAVITY

INNER ARM CABLES SET

EE-7347-001-301

TRACK CABLE SETS

EE-7347-001-215 - 15m
EE-7347-001-222 - 22m *
EE-7347-001-229 - 29m *
EE-7347-001-236 - 36m *

* REQUIRES NON -STANDARD PURGE AIR HOSE TO BE SPECIFIED IN PDL

R-30iB PLUS MATE CONTROLLER

AUXILIARY COMPONENT SIDEBOX

ENCODER / TRIGGER ARM HARNESS

MOTOR ARM HARNESS

AXIS 1

AXIS 2

AXIS 3

AXIS 4

AXIS 5

AXIS 6

JOINT #1

JOINT #2

JOINT #3

JOINT #4

PAINTER'S B ASE

TRIGGER VALVES

NOTE 1: NOT ALL CONTROLLER CONNECTIONS SHOWN.

NOTE 2: CABLES WITH -xxx PART NUMBERS INDICATE A LENGTH VARIANT.

NOTE 3: CABLES AND COMPONENTS SHOWN DOTTED ARE OPTIONAL. SEE PDL.

FANUC AMERICA CORPORATION

3900 W. Hamlin Rd., Rochester Hills, MI 48309-3253

P-40iA R-30iB MATE PLUS SINGLE ARM FLEX CABLE ROADMAP

SCALE

SHEET NO.

OF 1

OR AMEND NO.

EE-7347-002

9

ENGINEERING GUIDELINES

9.1 LOCTITE APPLICATION - EG-00008

Figure 9-1 EG-00008 Sheet 1 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION **ENGINEERING GUIDELINE**
Title: LOCTITE APPLICATION

Page 1 of 10
EG #: EG-00008

The purpose of this engineering guideline is to define the different Loctite products used at FAC and to provide directions for proper use.

This document is intended as a guideline only. Deviations are permitted and override this guideline when explicitly stated on engineering drawings/documentation.

1. Cleaning

Surfaces that are to have Loctite applied must be clean.

Acetone and Loctite "ODC-Free Cleaner & Degreaser" are approved cleaning agents.

Fasteners may be batch cleaned by submerging in acetone.

Critical areas where retaining compounds are used must be thoroughly cleaned to remove all preservative. Some preservatives are not visible and the parts may appear clean. Some preservatives contain nitrates which will stop Loctite from curing. All areas where retaining compound is used will have it's final cleaning with acetone.

The parts on which retaining compounds are used shall be thoroughly cleaned and "scrubbed" with acetone at least two times with a clean cloth. They may be washed beforehand in a solvent tank. The parts must then be wiped down and blown off before cleaning twice with acetone.

2. Primers

The engineering drawings shall call out primer when necessary.

Loctite 7649 (Primer N) ND 611 & Loctite 7471 (Primer T) ND 612

Loctite brand primers are used to ensure cure when applying Loctite threadlocker to two inactive metals. Primers will also speed up cure time.

Figure 9-3 EG-00008 Sheet 3 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION

ENGINEERING GUIDELINE

Title: LOCTITE APPLICATION

EG #: EG-00008

Page 3 of 10
EG #: EG-00008

Loctite 222 is a low strength threadlocker designed to lock and seal threaded fasteners, which require easy disassembly. Unless explicitly stated on engineering drawings/documentation, Loctite 222 is to be used on all carbon and stainless steel fasteners 5mm and smaller. The uncured appearance of this product is a purple liquid.

Loctite 242 ND 121 or 248(stick):

Loctite 242 is a medium strength treadlocker designed to lock and seal threaded fasteners that require normal disassembly with standard hand tools. Unless explicitly stated on engineering drawings/documentation, Loctite 242 is to be used on all carbon and stainless steel fasteners 6mm and above. The uncured appearance of this product is blue.

Loctite 262 ND 131 or 268(stick):

Loctite 262 is a high strength threadlocker designed to permanently lock and seal threaded fasteners. This product should not be applied unless explicitly stated on engineering drawings/documentation. The uncured appearance of this product is red.

Loctite 277 ND 146:

Loctite 277 is a high strength threadlocker designed to permanently lock and seal threaded fasteners. This product should not be applied unless explicitly stated on engineering drawings/documentation. The uncured appearance of this product is a red liquid.

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed.
- 2) Apply primer as required and allow to dry.
- 3) Select the specified product. Shake the liquid products before use.
- 4) **For Thru Holes**, apply several drops of the product onto the bolt at the nut engagement area.
- 5) **For Blind Holes**, the application method requires that product be applied to both parts. Apply several drops of the product down the internal threads to the bottom of the hole as well as the external treads of mating fastener. If product

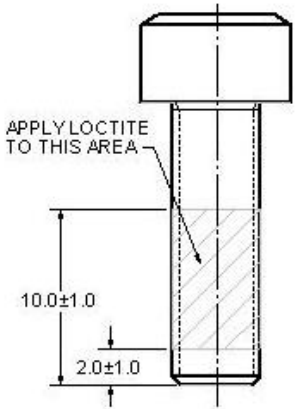
Figure 9-4 EG-00008 Sheet 4 of 10, LOCTITE APPLICATION

is only applied to the fastener, air pressure will force the liquid threadlocker to escape as the fastener is tightened down. Lack of uniform coverage creates air pockets, causing incomplete cures, which lead to failures.

6) Assemble and tighten as required. Fasteners must be torqued within 3 minutes.

3B. Thread Locking – Pre applied.

Loctite Dri Loc STS can be pre applied to fasteners provided the break away torque meets the FAC torque specification EG -00010.
For M5 fasteners it should be diluted 50 % by weight and applied as shown below.



4. Retaining

Loctite retaining compounds form a strong bond between press-fitted and slip-fitted parts. The product fills the inner space between components and cures to form a strong bond.

Loctite 680 is a high viscosity-retaining compound for joining fitted cylindrical parts. The product cures when confined in the absence of air between close fitting metals.

Figure 9-5 EG-00008 Sheet 5 of 10, LOCTITE APPLICATION

FANUC AMERICA CORPORATION

ENGINEERING GUIDELINE

Title: LOCTITE APPLICATION

Page 5 of 10

EG #: EG-00008

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed. Parts must be cleaned completely. Use acetone and scrub with a clean rag at least two times.
- 2) If parts are inactive metals or cure speed is too slow, spray with Loctite Primer 7471 or 7649 per the engineering drawing.
- 3) **For Slip Fitted Assemblies**, apply adhesive 360 degrees around OD and ID of mating parts. Use a rotating motion during assembly to ensure good coverage.
- 4) **For Press Fitted Assemblies**, apply adhesive 360 degrees around OD and ID of mating parts to insure the surfaces are completely coated during assembly. Assemble mating parts. When an undercut is provided, the adhesive shall be applied to the shaft to insure that the undercut is completely filled as the parts are pressed together.
- 5) Parts should not be disturbed until sufficient handling strength is achieved. Parts using Loctite 680 need to remain in fixture for 10 minutes. Full cure will be developed in 24 hours.

5a. Thread Sealant

Loctite “No More Leaks” is a thread sealant, which is designed to be used on tapered threads of plastic fittings that carry water or air. The sealant is solvent based and is non-curing so it shouldn’t be used for lines that carry paint or solvents. It is not a thread locker

Directions for use

- 1) Clean mating surfaces to ensure contamination has been removed.
- 2) Apply a 360-degree bead of sealant to the leading threads of the male fitting, leaving the first thread free. Force the sealant into the threads to ensure all voids are filled.
- 3) Assemble and tighten fittings as required.
- 4) Properly tightened fittings will seal instantly because it does not actually cure.

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5b. Thread Sealant and locker

A thread sealant will seal and secure metal and plastic pipes and fittings, filling the space between threaded metal parts, and hardening to prevent leakage. It can be used with primer on any metal and plastic fittings that are resistant to solvents such as Delrin (Acetal) or Nylon. The primer contains Acetone so it shouldn't be used on any plastic such as PVC that isn't resistant to paint solvents.

Loctite 567 (ND 460) is designed for locking and sealing metal tapered threads and fittings. The product cures when confined in the absence of air between close fitting metal surfaces and prevents loosening and leakage from shock and vibration. The product also provides lubrication, preventing galling between metal pipe threads and fittings.

Loctite 554 (ND 446) is similar Loctite 567 (ND 460). It should be used for locking and sealing of threads where maximum solvent resistance is required such as hoses and pipes that carry paint and paint solvents.

Directions for use

- 1) Clean mating surfaces to ensure contamination has been removed.
- 2) Apply primer to both the male and female threads as required and allow them to dry.
- 3) Apply a 360-degree bead of sealant to the leading threads of the male fitting, leaving the first thread free. Force the sealant into the threads to ensure all voids are filled.
- 4) Assemble and tighten fittings as required.
- 5) Properly tightened fittings will seal instantly and can withstand moderate pressure. For maximum pressure and solvent resistance allow the sealant to cure a minimum of 24 hours.

6. Gasketing

Gaskets are used to prevent fluid or gas leaks by forming impervious barriers that must remain intact and leak free for a prolonged period of time. Form-in-place gaskets are applied as a liquid sealant before parts are assembled. After

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assembly, the form-in-place gasket spreads and cures between assembled parts filling gaps, scratches and surface irregularities to provide a durable seal.

Loctite 518 ND 730 Gasket Eliminator is medium strength, solvent resistant flexible seal that will not tear or decay. The product cures when confined in the absence of air between close fitting metals. The product will fill gaps up to 0.25mm.

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed. The product gives best performance when surfaces are clean and free of grease.
- 2) Apply product in a continuous bead to one of the mating surfaces.
- 3) Assemble mating parts. Parts should be tightened as soon as possible after assembly.
- 4) Low pressure can be used to confirm a complete seal immediately after assembly. Allow time for full cure before applying high pressure.

7. Gasket Adhesives

Gasket adhesives are used to bond a gasket to its cover for ease of assembly.

Loctite "High Tack" is a flexible material that will bond gaskets to its cover. It can be used on all paper, rubber, cork or metal gaskets.

Directions for use

- 1) Clean mating surfaces to ensure all contamination has been removed.
- 2) Apply product with brush or spray.
- 3) Assemble gasket to cover. Parts should be allowed to cure for 5 minutes minimum before the cover/gasket is installed.

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- 4) After each cover removal, verify proper gasket attachment to the cover and reapply Permatex Hy-Tack HDWMO0000036300 if necessary.

8. List of the ND Industries part numbers – equivalent to Loctite.

Fanuc Item Number:	Current Item Description	New Item Description
HDWMO000001145O	LOCTITE #49550 ADHSV 1 OZBOTL	LOCTITE #49550 1oz / ND #39528
HDWMO000001146O	LOCTITE #242-41 SEALANT 250CC	LOCTITE #242-41 250cc/ND#12125
HDWMO000001147O ALT:HDWMO000002362O	LOCTITE #271-41 THRDLCKR 250CC	LOCTITE #271-41 250cc/ND#14025
HDWMO000001836O	LOCTITE #242-31 SEALANT 50CC	LOCTITE #242-31 50cc/ND #12150
HDWMO000002312O	LOCTITE #515-31 GSKT SEALANT	LOCTITE #515-31 50cc/ND #73250
HDWMO000002362O ALT:HDWMO000001147O	LOCTITE #271-31 SEALANT 50CC	LOCTITE #271-31 50cc/ND #14050
HDWMO000002450O	LOCTITE #26231 RETAIN COMPOUND	LOCTITE #26231 50cc/ND #13150
HDWMO000003076O	LOCTITE #680 RETAIN COMP 50CC	LOCTITE #680 50cc/ND #54150
HDWMO000003148O	LOCTITE #290-41 SEALANT 250CC	LOCTITE #290-31 50cc/ND#15050
HDWMO000003185O	LOCTITE #PST- 56747 SEALANT 50M	LOCTITE#PST56747 50ml/ND#46050
HDWMO000003186O	LOCTITE #22241 SEALANT 250CC	LOCTITE #22241 250cc/ND #11125

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HDWMO000003248O	LOCTITE #51831 SEALANT TUBE	LOCTITE #51831 50ml/ND #73050
HDWMO000003249O	LOCTITE #51845 SEALANT 300ML	LOCTITE #51845 300ml/ND#73025
HDWMO000003367O	LOCTITE #222-31 SEALANT 50 ML	LOCTITE #222-31 50ml/ND #11150
HDWMO000003658O	LOCTITE #21348 PRIMER N	LOCTITE #21348 PRI N/ND #61118
HDWMO0000060324O	LOCTITE #45440 PRISM 454 ADHES	LOCTITE #45440 20gm/ND #35420
HDWMO000056588O	LOCTITE #41650 416 INST ADHEV	LOCTITE #41650 1oz/ND #31628
HDWMO000057262O	Loctite #55441 Thread Sealant Locker	Loctite #55441 250cc ND #44625
HDWMO000057263O	ND #44650 50cc Same as Loctite #554	ND #44650 50cc

9.2 TORQUE SPECIFICATIONS - EG-00010

Figure 9-11 EG-00010 Sheet 1 of 5, TORQUE SPECIFICATION

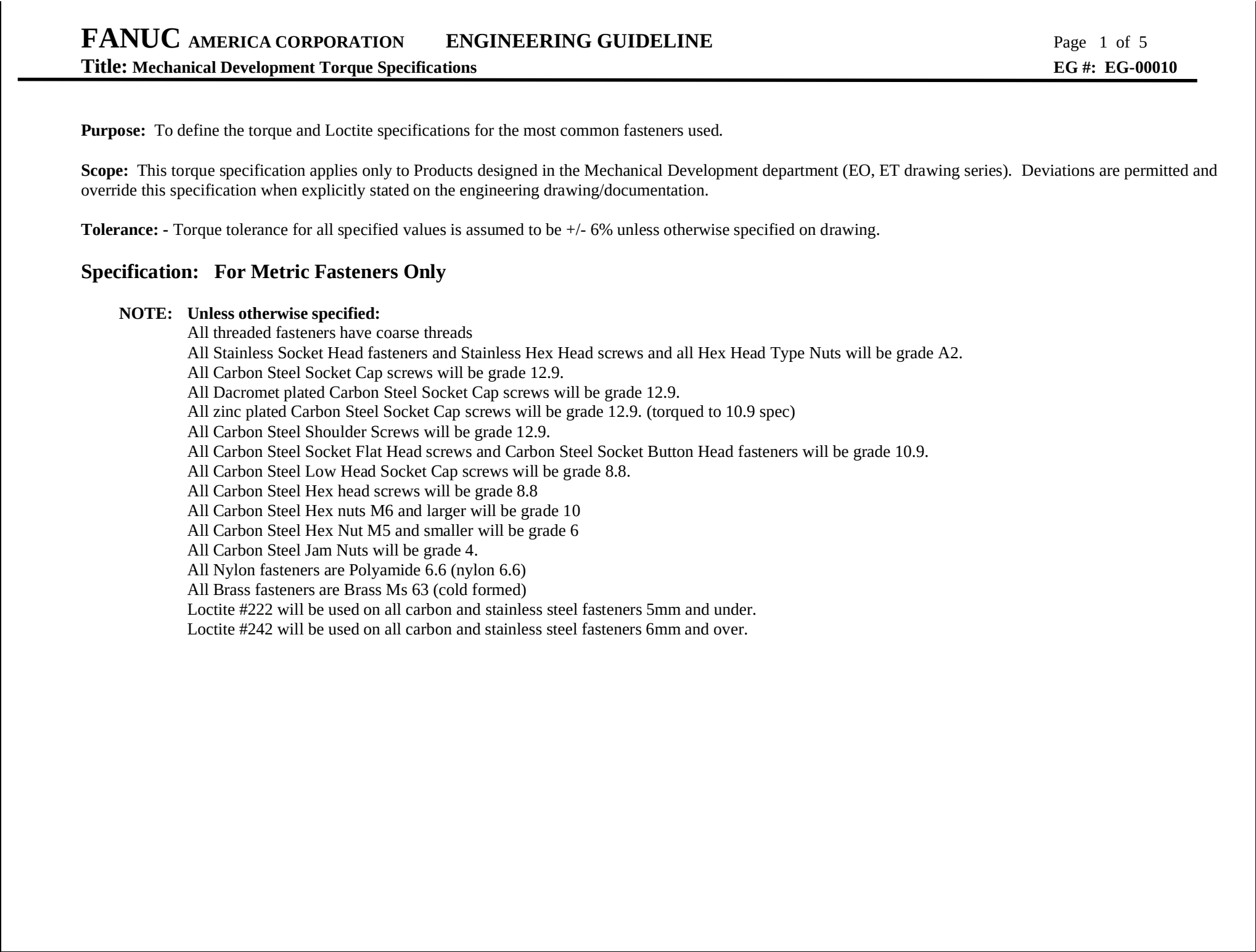


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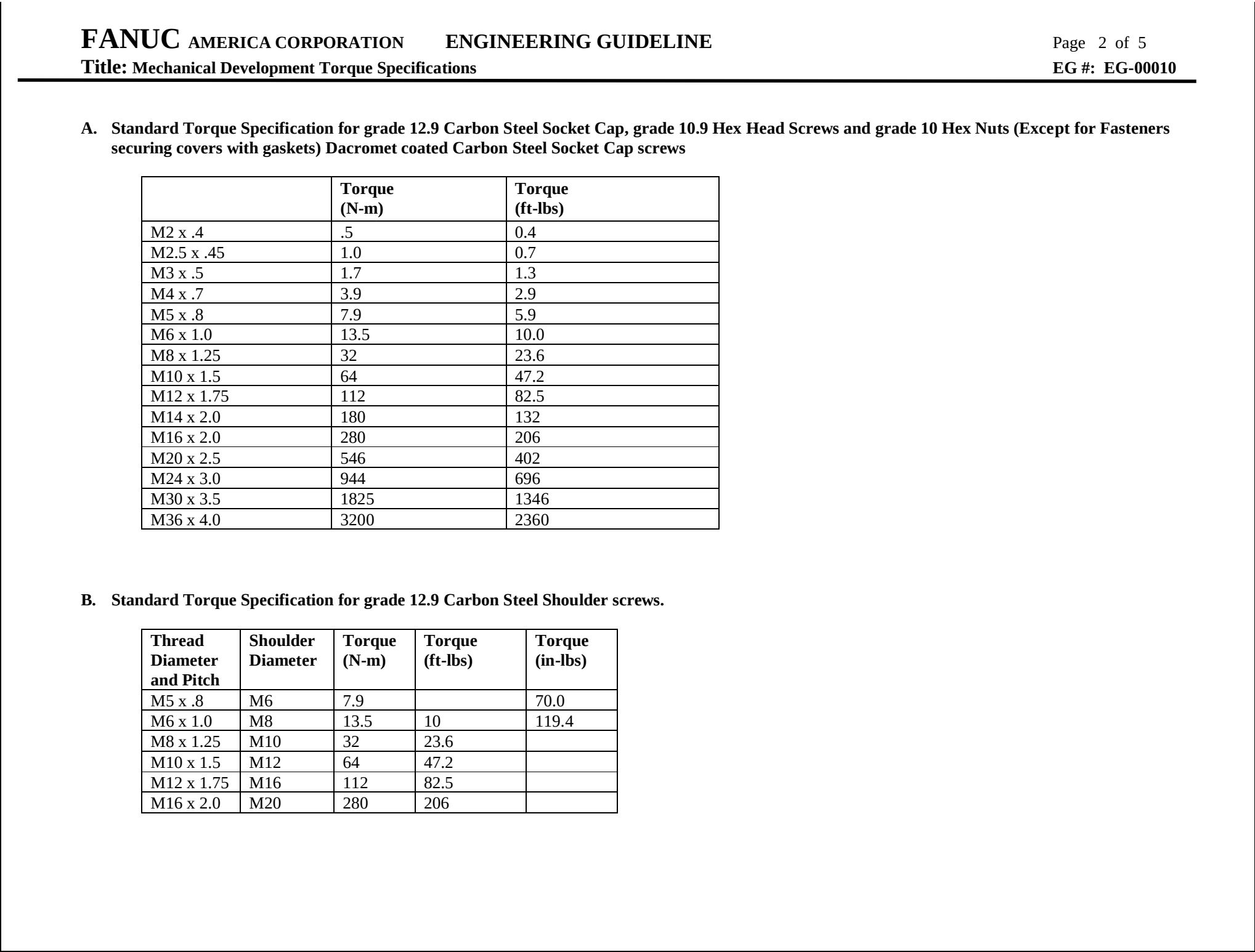


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C. Standard Torque Specification for all Stainless fasteners and Nuts (Except for Fasteners securing covers with gaskets)

	Torque (N-m)	Torque (ft-lbs)
M2 x .4	0.33	0.25
M2.5 x .45	0.66	0.5
M3 x .5	1.1	0.8
M4 x .7	2.6	2.0
M5 x .8	5.4	4.0
M6 x 1.0	9.2	6.8
M8 x 1.25	22.0	16.2
M10 x 1.5	43.5	32.0
M12 x 1.75	76.0	56.0
M14 x 2.0	121	89.0
M16 x 2.0	189	138
M20 x 2.5	368	271
M24 x 3.0	636	469
M30 x 3.5	1263	932
M36 x 4.0	2207	1628

D. Torque Specification for all Metric Carbon and Stainless Fasteners securing Covers with Gaskets

	Torque (N-m)	Torque (ft-lbs)
M4 x .7	2.0	1.5
M5 x .8	4.0	3.0
M6 x 1.0	6.3	4.6
M8 x 1.25	7.5	5.5
M10 x 1.5	8.4	6.2

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E. Standard Torque Specification for grade 8.8 Carbon Steel Hex Head screws, grade 10.9 Carbon steel Flat Head Socket Screws, grade 10.9 Button Head Socket Screws, grade 8.8 Carbon Steel Low Head Socket Cap Screws, grade 6 Carbon Steel Hex Nuts and all Carbon Steel Jam Nuts (Except for Fasteners securing covers with gaskets), grade 12.9 Zinc plated Carbon Steel Socket Cap screws

	Torque (N-m)	Torque (ft-lbs)
M2 x .4	0.4	0.3
M2.5 x .45	0.7	0.5
M3 x .5	1.0	0.7
M4 x .7	2.2	1.6
M5 x .8	4.3	3.1
M6 x 1.0	7.5	5.5
M8 x 1.25	18.0	13.2
M10 x 1.5	36.0	26.5
M12 x 1.75	67.0	49.4
M14 x 2.0	106.0	78.0
M16 x 2.0	161.0	118.5
M20 x 2.5	315.0	232
M24 x 3.0	544.0	401
M30 x 3.5	1080.0	796
M36 x 4.0	1890.0	1394

F. Standard Torque Specification for all Nylon fasteners and Nuts

	Torque (N-m)	Torque (ft-lbs)
M3 x .5	0.1	0.07
M4 x .7	0.2	0.15
M5 x .8	0.5	0.37
M6 x 1.0	1.0	0.74
M8 x 1.25	2.0	1.4
M10 x 1.5	3.0	2.2
M12 x 1.75	4.0	3.0
M14 x 2.0	6.0	4.4
M16 x 2.0	7.5	5.5

Figure 9-15 EG-00010 Sheet 5 of 5, TORQUE SPECIFICATION

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G. Standard Torque Specification for all Brass fasteners and Nuts

	Torque (N-m)	Torque (ft-lbs)
M2 x .4	0.14	0.10
M2.5 x .45	0.29	0.21
M3 x .5	0.5	0.37
M4 x .7	1.2	0.9
M5 x .8	2.2	1.6
M6 x 1.0	3.9	2.9
M8 x 1.25	9.0	6.6
M10 x 1.5	17.0	12.5